

INVERSE TRIGONOMETRIC FUNCTIONS



SYNOPSIS

➤ Inverse of a function :

$f : A \rightarrow B$ is bijective $\Leftrightarrow f^{-1} : B \rightarrow A$ exists and it is also bijective. All trigonometric functions are not bijective functions. By restricting the domains of the functions, we make them bijective

- The function $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$ defined by $f(x) = \sin x$ is a bijection.

Then $f^{-1} : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is also a bijection. This function is called **inverse sine** function and it is denoted by **Arc sine** x or $\text{Sin}^{-1} x$

- The function $f : [0, \pi] \rightarrow [-1, 1]$ defined by $f(x) = \cos x$ is a bijection.

Then $f^{-1} : [-1, 1] \rightarrow [0, \pi]$ is also a bijection. This function is called **inverse cosine** function and it is denoted by **Arc cos** x or $\text{Cos}^{-1} x$

- The function $f : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ defined by $f(x) = \tan x$ is a bijection.

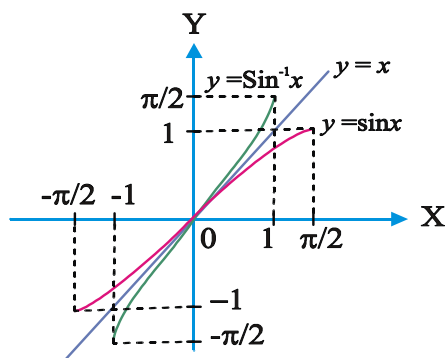
Then $f^{-1} : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ is also a bijection. This function is called **inverse tangent** function and it is denoted by **Arc tan** x or $\text{Tan}^{-1} x$

- **Domains and Ranges of Inverse trigonometric functions:**

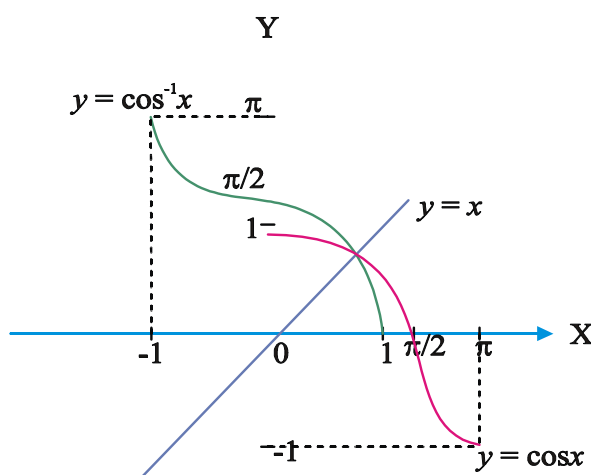
Function	Domain	Range
$\text{Sin}^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\text{Cos}^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\text{Tan}^{-1} x$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\text{Cot}^{-1} x$	$\mathbb{R}$	$(0, \pi)$
$\text{Sec}^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$
$\text{Cosec}^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$

➤ Graphs of inverse circular functions:

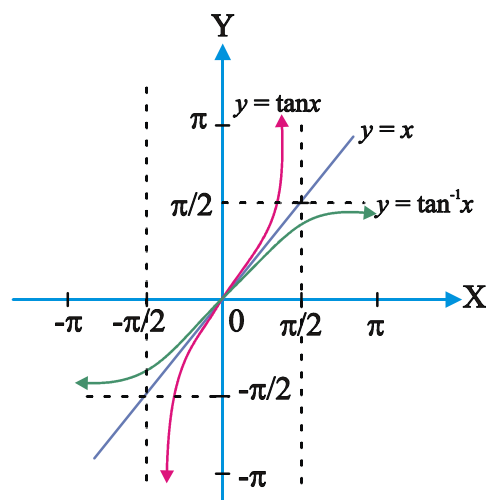
1. $y = \text{Sin}^{-1} x, |x| \leq 1, y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$



2. $y = \text{Cos}^{-1} x, |x| \leq 1, y \in [0, \pi]$

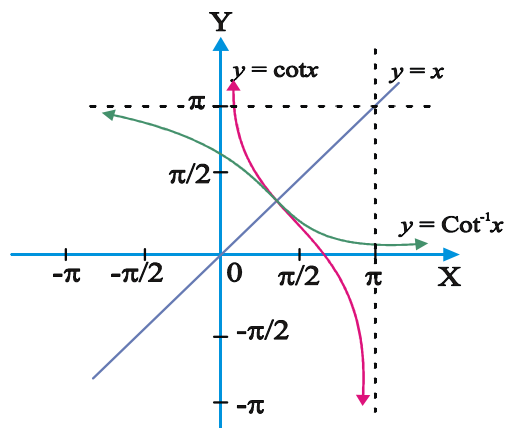


3. $y = \text{Tan}^{-1} x, x \in \mathbb{R}, y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

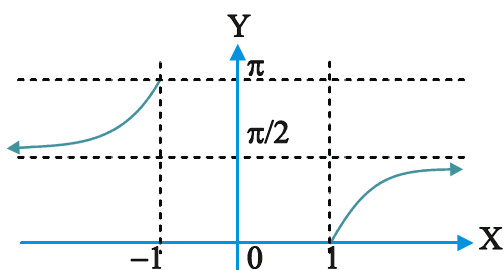


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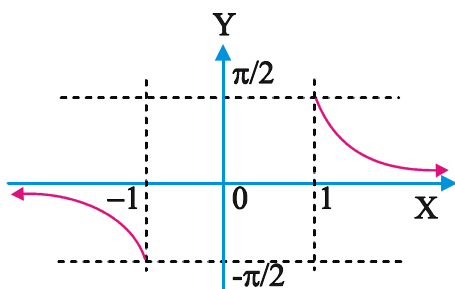
4. $y = \cot^{-1} x, x \in \mathbb{R}, y \in (0, \pi)$



5. $y = \sec^{-1} x, |x| \geq 1, y \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$



6. $y = \operatorname{cosec}^{-1} x, |x| \geq 1, y \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$



Properties of inverse trigonometric functions:

➤ i) $\sin^{-1} x = \operatorname{cosec}^{-1} \frac{1}{x}, \forall x \in [-1, 1], x \neq 0$

ii) $\cos^{-1} x = \sec^{-1} \frac{1}{x}, \forall x \in [-1, 1], x \neq 0$

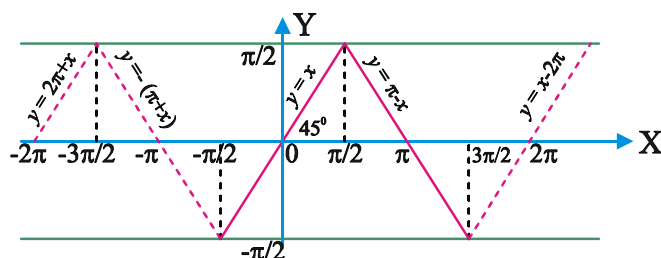
iii) $\tan^{-1} x = \begin{cases} \cot^{-1} \frac{1}{x}, & \forall x > 0 \\ -\pi + \cot^{-1} \frac{1}{x}, & \forall x < 0 \end{cases}$

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➤ Some useful periodic graphs:

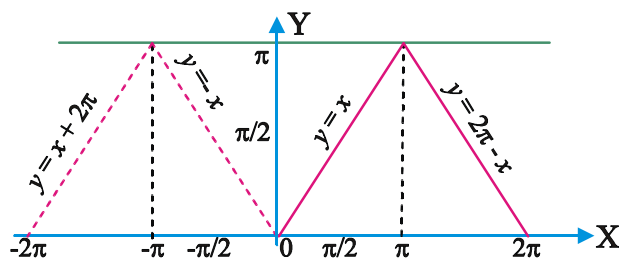
1. $y = \sin^{-1}(\sin x) = \begin{cases} -\pi - x, & -\frac{3\pi}{2} \leq x \leq -\frac{\pi}{2} \\ x, & -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \\ \pi - x, & \frac{\pi}{2} \leq x \leq \frac{3\pi}{2} \\ -2\pi + x, & \frac{3\pi}{2} \leq x \leq \frac{5\pi}{2} \end{cases}$

y is Periodic with period 2π and $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$



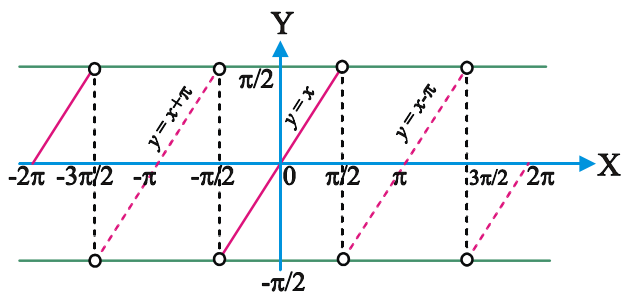
2. $y = \cos^{-1}(\cos x) = \begin{cases} -x, & -\pi \leq x \leq 0 \\ x, & 0 \leq x \leq \pi \\ 2\pi - x, & \pi \leq x \leq 2\pi \\ -2\pi + x, & 2\pi \leq x \leq 3\pi \end{cases}$

y is periodic with period 2π and $y \in [0, \pi]$

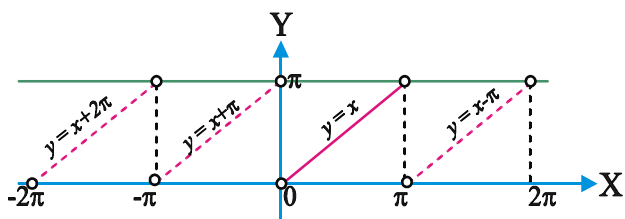


3. $y = \tan^{-1}(\tan x) = \begin{cases} x + \pi, & -\frac{3\pi}{2} < x < -\frac{\pi}{2} \\ x, & -\frac{\pi}{2} < x < \frac{\pi}{2} \\ -\pi + x, & \frac{\pi}{2} < x < \frac{3\pi}{2} \\ x - 2\pi, & \frac{3\pi}{2} < x < \frac{5\pi}{2} \end{cases}$

y is periodic with period π and $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

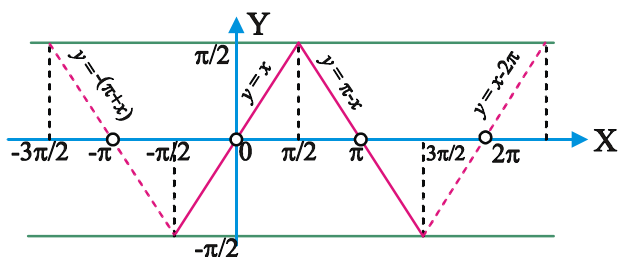


4. $y = \cot^{-1}(\cot x) = x$, $x \in (0, \pi)$ and so on.
 y is periodic with period π and $y \in (0, \pi)$



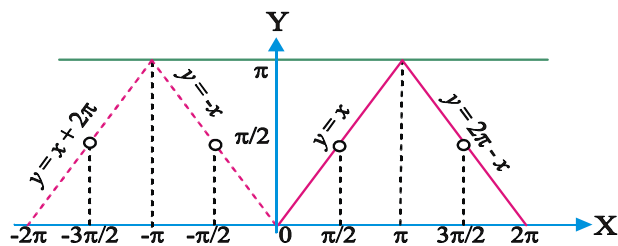
$$5. y = \operatorname{Cosec}^{-1}(\operatorname{cosec} x) = \begin{cases} x, & x \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right] \\ \pi - x, & x \in \left[\frac{\pi}{2}, \pi\right) \cup \left(\pi, \frac{3\pi}{2}\right] \\ \text{and so on} \end{cases}$$

y is periodic with period 2π and $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$



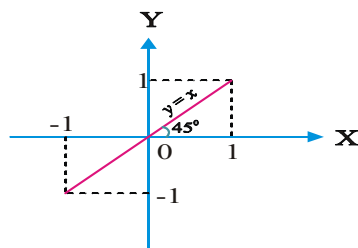
$$6. y = \sec^{-1}(\sec x) = \begin{cases} x, & x \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right] \\ 2\pi - x, & x \in \left[\pi, \frac{3\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right] \\ \text{and so on} \end{cases}$$

y is periodic with period 2π and $y \in [0, \pi]$

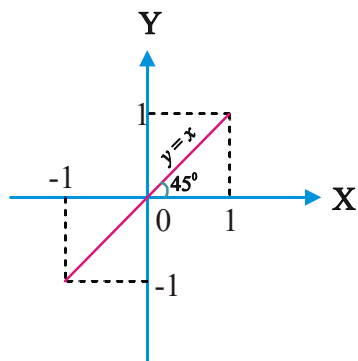


➤ **Some useful non-periodic graphs:**

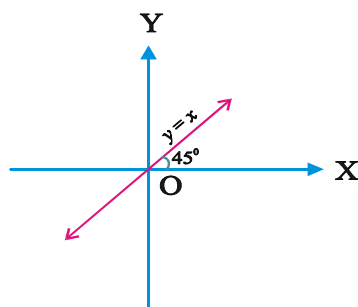
1. $y = \sin(\sin^{-1} x) = x$, $x \in [-1, 1]$, $y \in [-1, 1]$



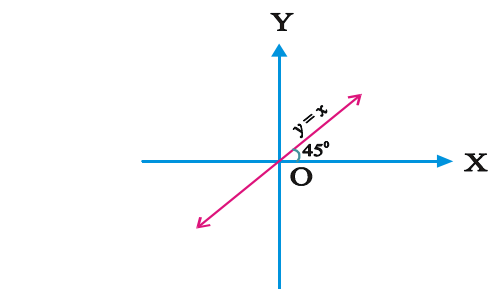
2. $y = \cos(\cos^{-1} x) = x$, $x \in [-1, 1]$, $y \in [-1, 1]$



3. $y = \tan(\tan^{-1} x) = x$, $x \in \mathbb{R}$, $y \in \mathbb{R}$

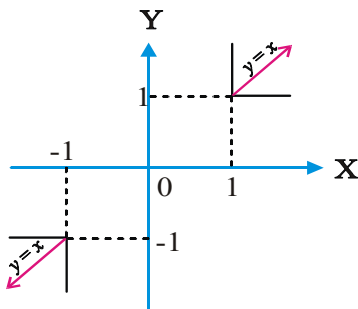


4. $y = \cot(\cot^{-1} x) = x$, $x \in \mathbb{R}$, $y \in \mathbb{R}$

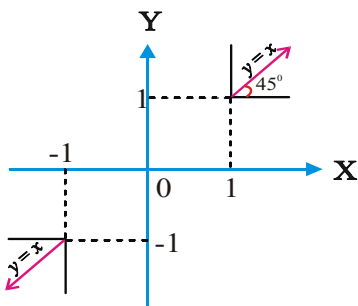


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5. $y = \operatorname{cosec}(\operatorname{cosec}^{-1}x) = x, |x| \geq 1, |y| \geq 1,$



6. $y = \sec(\sec^{-1}x) = x, |x| \geq 1, |y| \geq 1$



➤ Important Results:

- i) $\sin^{-1}(-x) = -\sin^{-1}x, \forall x \in [-1, 1]$
- ii) $\cos^{-1}(-x) = \pi - \cos^{-1}x, \forall x \in [-1, 1]$
- iii) $\tan^{-1}(-x) = -\tan^{-1}x, \forall x \in \mathbb{R}$
- iv) $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}x, \forall x \in \mathbb{R} - (-1, 1)$
- v) $\cot^{-1}(-x) = \pi - \cot^{-1}x, \forall x \in \mathbb{R}$
- vi) $\sec^{-1}(-x) = \pi - \sec^{-1}x, \forall x \in (-\infty, -1] \cup [1, \infty)$

W.E-1: $\cos^{-1}\left(\frac{-1}{2}\right) - 2\sin^{-1}\left(\frac{1}{2}\right) + 3\cos^{-1}\left(\frac{-1}{\sqrt{2}}\right) - 4\tan^{-1}(-1) =$ (EAM-2009)

Sol: $\cos^{-1}\left(\frac{-1}{2}\right) - 2\sin^{-1}\left(\frac{1}{2}\right) + 3\cos^{-1}\left(\frac{-1}{\sqrt{2}}\right) - 4\tan^{-1}(-1) = \pi - \cos^{-1}\left(\frac{1}{2}\right) - 2\left(\frac{\pi}{6}\right) + 3\left(\pi - \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)\right) + 4\tan^{-1}(1)$
 $= \pi - \frac{\pi}{3} - \frac{\pi}{3} + 3\left(\pi - \frac{\pi}{4}\right) + 4\left(\frac{\pi}{4}\right)$
 $= \frac{\pi}{3} + 3\left(\frac{3\pi}{4}\right) + \pi = \frac{43\pi}{12}$

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- i) $\theta \in [0, \pi]$ then $\sin^{-1}(\cos \theta) = \frac{\pi}{2} - \theta$
- ii) $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ then $\cos^{-1}(\sin \theta) = \frac{\pi}{2} - \theta$
- iii) $\theta \in (0, \pi)$ then $\tan^{-1}(\cot \theta) = \frac{\pi}{2} - \theta$
- iv) $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ then $\cot^{-1}(\tan \theta) = \frac{\pi}{2} - \theta$
- v) $\theta \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$ then

$$\sec^{-1}(\operatorname{cosec} \theta) = \frac{\pi}{2} - \theta$$

- vi) $\theta \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$ then

$$\operatorname{cosec}^{-1}(\sec \theta) = \frac{\pi}{2} - \theta$$

- i) $\sin^{-1}x = \begin{cases} \cos^{-1}\sqrt{1-x^2} & \text{if } 0 \leq x \leq 1 \\ -\cos^{-1}\sqrt{1-x^2} & \text{if } -1 \leq x < 0 \\ \tan^{-1}\frac{x}{\sqrt{1-x^2}} & \text{if } x \in (-1, 1) \end{cases}$
- ii) $\cos^{-1}x = \begin{cases} \sin^{-1}\sqrt{1-x^2} & \text{if } x \in [0, 1] \\ \pi - \sin^{-1}\sqrt{1-x^2} & \text{if } x \in [-1, 0) \end{cases}$
- iii) $\tan^{-1}x = \begin{cases} \sin^{-1}\frac{x}{\sqrt{1+x^2}} & \text{for } x > 0 \\ \cos^{-1}\frac{1}{\sqrt{1+x^2}} & \text{for } x < 0 \end{cases}$

W.E-2: The value of x , where $x > 0$ and

$\tan\left(\sec^{-1}\frac{1}{x}\right) = \sin(\tan^{-1}2)$ is (EAM-2007)

Sol: $\tan\left(\sec^{-1}\frac{1}{x}\right) = \sin(\tan^{-1}2)$

$$\tan\left(\tan^{-1}\frac{\sqrt{1-x^2}}{x}\right) = \sin\left(\sin^{-1}\frac{2}{\sqrt{1+2^2}}\right)$$

$$\left(\because \tan^{-1}x = \sin^{-1}\frac{x}{\sqrt{1+x^2}}\right)$$

$$\frac{\sqrt{1-x^2}}{x} = \frac{2}{\sqrt{5}} \Rightarrow 5(1-x^2) = 4x^2$$

$$\Rightarrow x^2 = \frac{5}{9} \Rightarrow x = \frac{\sqrt{5}}{3}$$

- i) $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \forall x \in [-1, 1]$
 ii) $\tan^{-1} x + \cot^{-1} x = \pi/2, \forall x \in \mathbb{R}$
 iii) $\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}, \forall x \in (-\infty, -1] \cup [1, \infty)$

$$\sin^{-1} x + \sin^{-1} y = \begin{cases} \sin^{-1} \{x\sqrt{1-y^2} + y\sqrt{1-x^2}\} \\ 0 \leq x, y, x^2 + y^2 \leq 1 \text{ or} \\ -1 \leq x, y \leq 1, xy < 0 \text{ and } x^2 + y^2 > 1 \\ \pi - \sin^{-1} \{x\sqrt{1-y^2} + y\sqrt{1-x^2}\} \\ \text{if } 0 < x, y \leq 1 \text{ and } x^2 + y^2 > 1 \\ -\pi - \sin^{-1} \{x\sqrt{1-y^2} + y\sqrt{1-x^2}\} \\ \text{if } -1 \leq x, y < 0 \text{ and } x^2 + y^2 > 1 \end{cases}$$

$$\sin^{-1} x - \sin^{-1} y = \begin{cases} \sin^{-1} \{x\sqrt{1-y^2} - y\sqrt{1-x^2}\} \\ 0 \leq x, y < 1 \text{ and } x^2 + y^2 > 1 \text{ or} \\ -1 \leq x, y \leq 1, xy < 0 \text{ and } x^2 + y^2 \leq 1 \\ \pi - \sin^{-1} \{x\sqrt{1-y^2} - y\sqrt{1-x^2}\} \\ \text{if } 0 < x \leq 1, -1 \leq y < 1 \text{ and } x^2 + y^2 > 1 \\ -\pi - \sin^{-1} \{x\sqrt{1-y^2} - y\sqrt{1-x^2}\} \\ \text{if } 0 < y \leq 1, -1 \leq x < 0 \text{ and } x^2 + y^2 > 1 \end{cases}$$

$$\cos^{-1} x + \cos^{-1} y = \begin{cases} \cos^{-1} \{xy - \sqrt{1-x^2}\sqrt{1-y^2}\} \\ \text{if } -1 \leq x, y \leq 1 \text{ and } x+y \geq 0 \\ 2\pi - \cos^{-1} \{xy - \sqrt{1-x^2}\sqrt{1-y^2}\} \\ \text{if } -1 \leq x, y \leq 1 \text{ and } x+y < 0 \end{cases}$$

$$\cos^{-1} x - \cos^{-1} y = \begin{cases} \cos^{-1} \{xy + \sqrt{1-x^2}\sqrt{1-y^2}\} \\ \text{if } -1 \leq x, y \leq 1, x \leq y \\ -\cos^{-1} \{xy + \sqrt{1-x^2}\sqrt{1-y^2}\} \\ \text{if } -1 \leq y \leq 0, 0 < x \leq 1 \text{ and } x \geq y \end{cases}$$

W.E-3: If $\frac{1}{2} \leq x \leq 1$ then

$\cos^{-1} \left(\frac{x}{2} + \frac{1}{2} \sqrt{3-3x^2} \right) + \cos^{-1} x$ is equal to
(EAM-2012)

Sol: $\cos^{-1} x + \cos^{-1} \left(\frac{x}{2} + \frac{1}{2} \sqrt{3-3x^2} \right)$
 $= \cos^{-1} x + \cos^{-1} \left(x \cdot \frac{1}{2} + \frac{\sqrt{3}}{2} \sqrt{1-x^2} \right)$
 $= \cos^{-1} x + \cos^{-1} \left(\frac{1}{2} \right) - \cos^{-1} x = \cos^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{3}$

$$\tan^{-1} x + \tan^{-1} y = \begin{cases} \tan^{-1} \left(\frac{x+y}{1-xy} \right) & \text{if } x > 0, y > 0, xy < 1 \\ \pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right) & \text{if } x > 0, y > 0, xy > 1 \\ -\pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right) & \text{if } x < 0, y < 0, xy > 1 \\ \frac{\pi}{2}, & \text{if } xy = 1 \end{cases}$$

$$\tan^{-1} x - \tan^{-1} y = \begin{cases} \tan^{-1} \left(\frac{x-y}{1+xy} \right) & \text{for } xy > -1 \\ \pi + \tan^{-1} \left(\frac{x-y}{1+xy} \right) & \text{if } x > 0, y < 0 \text{ and } xy < -1 \\ -\pi + \tan^{-1} \left(\frac{x-y}{1+xy} \right) & \text{if } x < 0, y > 0 \text{ and } xy < -1 \end{cases}$$

- (i) If x, y, z have same sign and $xy + yz + zx < 1$ then

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \tan^{-1} \left(\frac{x+y+z-xyz}{1-xy-yz-zx} \right)$$

- (ii) $\tan^{-1} x_1 + \tan^{-1} x_2 + \dots + \tan^{-1} x_n =$

$$\tan^{-1} \left(\frac{S_1 - S_3 + S_5 - \dots}{1 - S_2 + S_4 - S_6 + \dots} \right)$$

where S_1 = sum of values, S_2 = sum of product of taken two elements at a time and so on.,

S_n = product of values.

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W.E-4: The value of $\text{Cot}\left(\text{Cosec}^{-1}\frac{5}{3} + \text{Tan}^{-1}\frac{2}{3}\right) =$
(AIE-2008)

Sol: $\text{Cot}\left(\text{Cosec}^{-1}\frac{5}{3} + \text{Tan}^{-1}\frac{2}{3}\right) = \text{Cot}\left(\text{Sin}^{-1}\frac{3}{5} + \text{Tan}^{-1}\frac{2}{3}\right)$

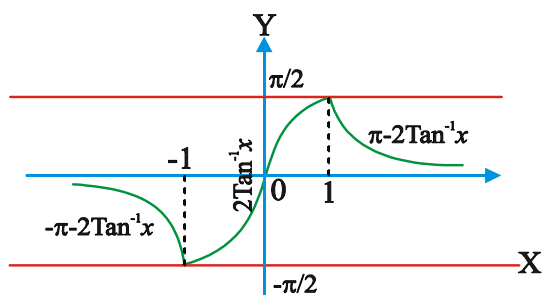
$$= \text{Cot}\left(\text{Tan}^{-1}\frac{3}{4} + \text{Tan}^{-1}\frac{2}{3}\right) \left[\because \text{Sin}^{-1}x = \text{Tan}^{-1}\frac{x}{\sqrt{1-x^2}} \right]$$

$$= \text{Cot}\left[\text{Tan}^{-1}\left(\frac{\frac{3}{4} + \frac{2}{3}}{1 - \left(\frac{3}{4}\right)\left(\frac{2}{3}\right)}\right)\right] = \text{Cot}\left[\text{Tan}^{-1}\left(\frac{17}{12}\right)\right]$$

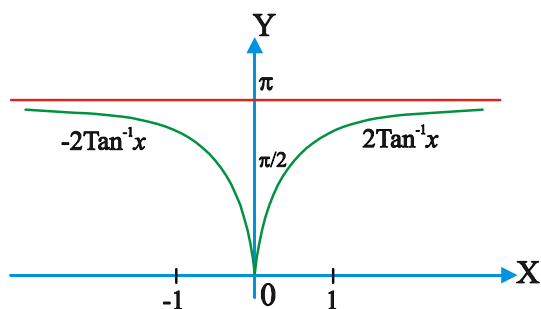
$$= \cot \cot^{-1}\left(\frac{17}{12}\right) = \cot \tan^{-1}\left(\frac{17}{12}\right) = \frac{12}{17}$$

➤ **Transformation of Inverse functions by elementary substitution and their graphs:**

i) $\text{Sin}^{-1}\frac{2x}{1+x^2} = \begin{cases} -\pi - 2\text{Tan}^{-1}x & x \leq -1 \\ 2\text{Tan}^{-1}x & -1 \leq x \leq 1 \\ \pi - 2\text{Tan}^{-1}x & x \geq 1 \end{cases}$

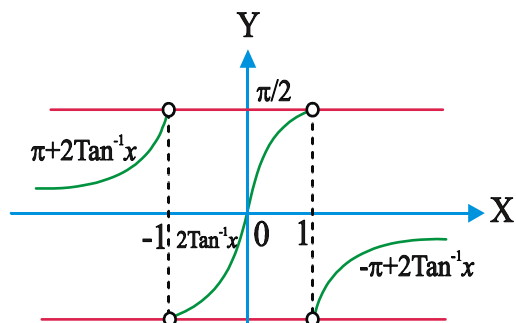


ii) $\text{Cos}^{-1}\frac{1-x^2}{1+x^2} = \begin{cases} 2\text{Tan}^{-1}x & x \geq 0 \\ -2\text{Tan}^{-1}x & x < 0 \end{cases}$

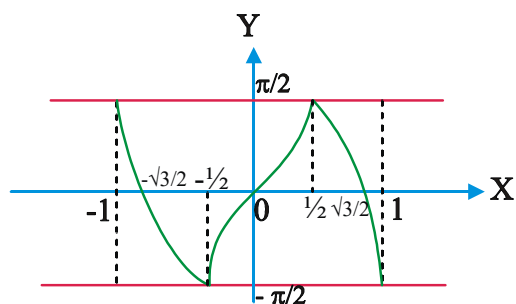


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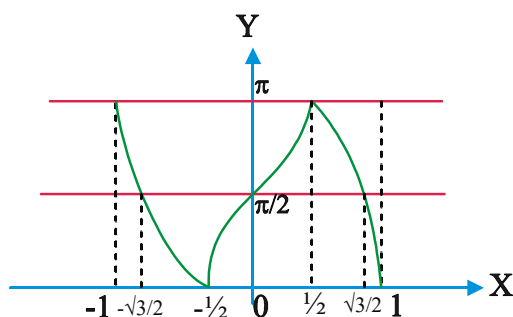
iii) $\text{Tan}^{-1}\frac{2x}{1-x^2} = \begin{cases} \pi + 2\text{Tan}^{-1}x & x < -1 \\ 2\text{Tan}^{-1}x & -1 < x < 1 \\ -\pi + 2\text{Tan}^{-1}x & x > 1 \end{cases}$



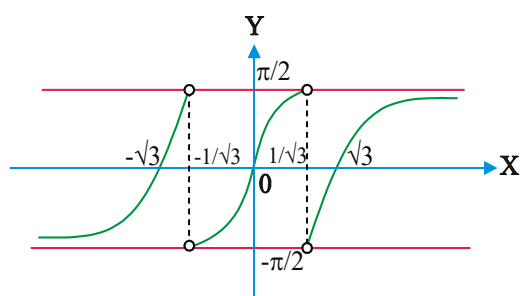
➤ i) $\text{Sin}^{-1}(3x-4x^3) = \begin{cases} -(\pi + 3\text{Sin}^{-1}x) & \text{if } -1 \leq x \leq -1/2 \\ 3\text{Sin}^{-1}x & \text{if } -1/2 \leq x \leq 1/2 \\ \pi - 3\text{Sin}^{-1}x & \text{if } 1/2 \leq x \leq 1 \end{cases}$



ii) $\text{Cos}^{-1}(4x^3-3x) = \begin{cases} 3\text{Cos}^{-1}x - 2\pi & \text{if } -1 \leq x \leq -1/2 \\ 2\pi - 3\text{Cos}^{-1}x & \text{if } -1/2 \leq x \leq 1/2 \\ 3\text{Cos}^{-1}x & \text{if } 1/2 \leq x \leq 1 \end{cases}$



$$\text{iii) } \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right) = \begin{cases} 3\tan^{-1}x & \text{if } -\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}} \\ \pi + 3\tan^{-1}x & \text{if } x > \frac{1}{\sqrt{3}} \\ -\pi + 3\tan^{-1}x & \text{if } x < -\frac{1}{\sqrt{3}} \end{cases}$$



$$\text{➤ i) } 2\sin^{-1}x = \begin{cases} -\pi - \sin^{-1}(2x\sqrt{1-x^2}) & \text{if } -1 \leq x \leq -\frac{1}{\sqrt{2}} \\ \sin^{-1}(2x\sqrt{1-x^2}) & \text{if } -\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}} \\ \pi - \sin^{-1}(2x\sqrt{1-x^2}) & \text{if } \frac{1}{\sqrt{2}} \leq x \leq 1 \end{cases}$$

$$\text{ii) } 2\cos^{-1}x = \begin{cases} \cos^{-1}(2x^2-1) & \text{if } 0 \leq x \leq 1 \\ 2\pi - \cos^{-1}(2x^2-1) & \text{if } -1 \leq x \leq 0 \end{cases}$$

$$\text{iii) } 2\tan^{-1}x = \begin{cases} \tan^{-1}\left(\frac{2x}{1-x^2}\right) & \text{if } -1 < x < 1 \\ \pi + \tan^{-1}\left(\frac{2x}{1-x^2}\right) & \text{if } x > 1 \\ -\pi + \tan^{-1}\left(\frac{2x}{1-x^2}\right) & \text{if } x < -1 \end{cases}$$

$$\text{iv) } 2\tan^{-1}x = \begin{cases} -\pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right) & \text{if } x < -1 \\ \sin^{-1}\left(\frac{2x}{1+x^2}\right) & \text{if } -1 \leq x \leq 1 \\ \pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right) & \text{if } x > 1 \end{cases}$$

$$\text{v) } 2\tan^{-1}x = \begin{cases} -\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) & \text{if } -\infty < x \leq 0 \\ \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) & \text{if } 0 \leq x < \infty \end{cases}$$

W.E-5: $\sin^{-1}\frac{4}{5} + 2\tan^{-1}\frac{1}{3} =$ (EAM-05)

Sol: $\sin^{-1}\frac{4}{5} + 2\tan^{-1}\frac{1}{3}$

$$= \sin^{-1}\frac{4}{5} + \tan^{-1}\left(\frac{2\left(\frac{1}{3}\right)}{1-\left(\frac{1}{3}\right)^2}\right) \left[\because 2\tan^{-1}x = \tan^{-1}\left(\frac{2x}{1-x^2}\right) \right]$$

$$= \sin^{-1}\frac{4}{5} + \tan^{-1}\frac{3}{4}$$

$$= \sin^{-1}\frac{4}{5} + \cos^{-1}\frac{4}{5} \left[\because \tan^{-1}x = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right) \right]$$

$$= \frac{\pi}{2} \left[\because \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2} \right]$$

➤ **Some important facts:**

1) $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \frac{\pi}{2}$, if $xy + yz + zx = 1$

2) $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$,
if $x + y + z = xyz$ (EAM-2014)

3) $\tan^{-1}\frac{a}{x} + \tan^{-1}\frac{b}{x} = \frac{\pi}{2}$, then $x = \sqrt{ab}$

4) $\sin^{-1}\frac{a}{x} + \sin^{-1}\frac{b}{x} = \frac{\pi}{2}$, then $x = \sqrt{a^2 + b^2}$

5) $\tan^{-1}\left(\frac{p}{q}\right) + \tan^{-1}\left(\frac{q-p}{q+p}\right) = \frac{\pi}{4}$

6) $\tan^{-1}\left(\frac{1+x}{1-x}\right) = \frac{\pi}{4} + \tan^{-1}x$ if $x < 1$

7) $\tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{\pi}{4} - \tan^{-1}x$ if $x > -1$

8) If $\tan^{-1}x + \tan^{-1}y = \frac{\pi}{2}$ then $xy = 1$.

9) $\cot^{-1}x + \cot^{-1}y = \frac{\pi}{2}$ then $xy = 1$

10) If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = 3\pi$
then $xy + yx + zx = 3$

11) If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \frac{3\pi}{2}$
then $xy + yx + zx = 3$

INVERSE TRIGONOMETRIC FUNCTIONS

12) If $\sin^{-1}x + \sin^{-1}y = \theta$ then $\cos^{-1}x + \cos^{-1}y = \pi - \theta$

13) If $\cos^{-1}x + \cos^{-1}y = \theta$ then $\sin^{-1}x + \sin^{-1}y = \pi - \theta$

14) If $a\sin^{-1}x - b\cos^{-1}x = c$ then

$$a\sin^{-1}x + b\cos^{-1}x = \frac{\pi ab + c(a-b)}{a+b}$$

15) If $\cos^{-1}\frac{x}{a} + \cos^{-1}\frac{y}{b} = \theta$ then

$$\frac{x^2}{a^2} - \frac{2xy}{ab}\cos\theta + \frac{y^2}{b^2} = \sin^2\theta$$

16) If $\sin^{-1}\frac{x}{a} + \sin^{-1}\frac{y}{b} = \theta$ then

$$\frac{x^2}{a^2} + \frac{2xy}{ab}\cos\theta + \frac{y^2}{b^2} = \sin^2\theta$$

17) $\tan^{-1}\left(\frac{1}{1+x(x+1)}\right) + \tan^{-1}\left(\frac{1}{1+(x+1)(x+2)}\right) + \dots +$

$$\tan^{-1}\left(\frac{1}{1+(x+n-1)(x+n)}\right) = \tan^{-1}(x+n) - \tan^{-1}x, n \in \mathbb{N}$$

➤ If an expression contains

i) $\sqrt{a^2 - x^2}$, put $x = a \sin \theta, \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

ii) $\sqrt{a^2 + x^2}$, put $x = a \tan \theta, \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

iii) $\sqrt{x^2 - a^2}$, put $x = a \sec \theta, \theta \in (0, \pi)$

➤ **Range of some special inverse Trigonometric Functions :**

i) $\frac{\pi^3}{32} \leq (\sin^{-1}x)^3 + (\cos^{-1}x)^3 \leq \frac{7\pi^3}{8}$

ii) $\frac{\pi^2}{8} \leq (\sin^{-1}x)^2 + (\cos^{-1}x)^2 \leq \frac{5\pi^2}{4}$

iii) $-\frac{\pi^2}{4} \leq (\cos^{-1}x)^2 - (\sin^{-1}x)^2 \leq \frac{3\pi^2}{4}$

W.E-6 : The set of values of x such that $\sin^{-1}x - \cos^{-1}x > 0$ are

Sol: $\sin^{-1}x - \cos^{-1}x > 0 \Rightarrow \sin^{-1}x > \frac{\pi}{4}$

$$\therefore x \in \left[\frac{1}{\sqrt{2}}, 1\right] \text{ as } |x| \leq 1$$

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W.E-7: The sum to the n terms of the series

$$\operatorname{Cosec}^{-1}\sqrt{10} + \operatorname{Cosec}^{-1}\sqrt{50} + \operatorname{Cosec}^{-1}\sqrt{70} + \dots$$

$$+ \dots + \operatorname{Cosec}^{-1}\sqrt{(n^2+1)(n^2+2n+2)} \text{ is}$$

Sol: Let $y = \operatorname{Cosec}^{-1}\sqrt{(n^2+1)(n^2+2n+2)}$

$$\operatorname{Cosec}^2 y = (n^2+1) \left[(n^2+1) + 2n+1 \right]$$

$$= (n^2+1)^2 + 2n(n^2+1) + n^2 + 1$$

$$= (n^2+n+1)^2 + 1$$

$$\cot^2 y = (n^2+n+1)^2 \left[\because \operatorname{cosec}^2 \theta - \cot^2 \theta = 1 \right]$$

$$\tan y = \frac{1}{n^2+n+1} = \frac{(n+1)-n}{1+(n+1)n}$$

$$y = \tan^{-1} \left[\frac{(n+1)-n}{1+(n+1)n} \right]$$

$$y = \tan^{-1}(n+1) - \tan^{-1}n$$

Thus sum of n terms of the given series

$$\begin{aligned} y &= (\tan^{-1}2 - \tan^{-1}1) + (\tan^{-1}3 - \tan^{-1}2) + \\ &(\tan^{-1}4 - \tan^{-1}3) + \dots + (\tan^{-1}(n+1) - \tan^{-1}n) \\ &= \tan^{-1}(n+1) - \tan^{-1}1 = \tan^{-1}(n+1) - \frac{\pi}{4} \end{aligned}$$

W.E-8: $\sin^{-1}(\sin 5) =$

Sol: Here $\theta = 5$ rad, Clearly it does not lie between

$-\frac{\pi}{2}$ and $\frac{\pi}{2}$. But $2\pi - 5$ and $5 - 2\pi$ both lies

between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$.

$$\sin(5 - 2\pi) = \sin[-(2\pi - 5)] = -\sin(2\pi - 5) = \sin 5$$

$$\therefore \sin^{-1}(\sin 5) = \sin^{-1}(\sin(5 - 2\pi)) = 5 - 2\pi$$

W.E-9 : $\cos^{-1}(\cos 10)$ is equal to....

Sol: We know that $\cos^{-1}(\cos \theta) = \theta$ if $0 \leq \theta \leq \pi$

Here $\theta = 10$ rad, clearly, it does not lie between 0 and π . But $4\pi - 10$ lies between 0 and π

$$\therefore \cos^{-1}(\cos 10) = \cos^{-1}(\cos(4\pi - 10)) = 4\pi - 10$$

W.E-10: $\tan^{-1}(\tan(-6))$ is equal to....

Sol: We know that $\tan^{-1}(\tan\theta) = \theta$ if $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

Here $\theta = -6 \text{ rad}$, does not lie between

$-\frac{\pi}{2}$ and $\frac{\pi}{2}$

But $2\pi - 6$ lies between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$

Now $\tan(2\pi - 6) = -\tan 6 = \tan(-6)$

$\therefore \tan^{-1}(\tan(-6)) = \tan^{-1}(\tan(2\pi - 6)) = 2\pi - 6$



C.U.Q

1. $\tan(\cos^{-1}x) =$

- 1) $\frac{\sqrt{1-x^2}}{x}$ 2) $\frac{x}{1-x^2}$ 3) $\frac{\sqrt{1+x^2}}{x}$ 4) $\sqrt{1-x^2}$

2. The value of $\sin(\cot^{-1}(\cos(\tan^{-1}x)))$ is

- 1) $\sqrt{\frac{x^2+2}{x^2+1}}$ 2) $\sqrt{\frac{x^2+1}{x^2+2}}$
3) $\frac{x}{\sqrt{x^2+2}}$ 4) $\frac{1}{\sqrt{x^2+2}}$

3. $\tan[2\tan^{-1}(\cos x)] =$

- 1) $2\tan x \cos x$ 2) $2\tan x \operatorname{cosec} x$
3) $2\cot x \operatorname{cosec} x$ 4) $2\sec x$

4. $\cos^2[\tan^{-1}(\sin(\cot^{-1}x))] =$

- 1) $\frac{x^2-1}{x^2+2}$ 2) $\frac{x^2+1}{x^2-2}$ 3) $\frac{x^2+1}{x^2+2}$ 4) $\frac{x^2-1}{x^2+1}$

5. If $a > b > c > 0$, then

$$\cot^{-1}\left(\frac{ab+1}{a-b}\right) + \cot^{-1}\left(\frac{bc+1}{b-c}\right) + \cot^{-1}\left(\frac{ca+1}{c-a}\right)$$

- 1) 0 2) $\pi/2$ 3) π 4) $3\pi/2$

6. If $\sec^{-1}\sqrt{1+x^2} + \operatorname{cosec}^{-1}\frac{\sqrt{1+y^2}}{y} + \cot^{-1}\frac{1}{z} = \pi$,

then

- 1) $x + y + z = 0$ 2) $x + y + z = 1$
3) $x + y + z = xyz$ 4) $x + y + z = -xyz$

7. $\tan\left[\frac{1}{2}\sin^{-1}\left(\frac{2x}{1+x^2}\right) - \frac{1}{2}\cos^{-1}\left(\frac{1-y^2}{1+y^2}\right)\right] =$

- 1) 0 2) 1 3) $\frac{x-y}{1+xy}$ 4) $\frac{2x}{1-x^2}$

8. $\cos[\sin^{-1}(2\cos^2\theta - 1) + \cos^{-1}(1 - 2\sin^2\theta)] =$

- 1) 0 2) 1 3) -1 4) $\frac{\pi}{2}$

9. If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \frac{\pi}{2}$ then

(Eam-2010)

- 1) $xy + yz + zx = 1$ 2) $x^2 + y^2 + z^2 + 2xyz = 1$
3) $x + y + z = xyz$ 4) $\sum x + \sum yz = 1 + xyz$

10. $\cos^{-1}(x/a) + \cos^{-1}(y/b) = \theta$ then

$$\frac{x^2}{a^2} - \frac{2xy}{ab} \cos \theta + \frac{y^2}{b^2} =$$

- 1) $\sin^2\theta$ 2) $\cos^2\theta$ 3) $\tan^2\theta$ 4) $\cot^2\theta$

11. $2\sin^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$ holds good for

- 1) $x \in [0, 1]$ 2) $x \in [-1, 1]$
3) $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$ 4) $x \in [-1, 0]$

12. $\operatorname{cosec}^{-1}(\cos x)$ is defined if

- 1) $x \in [-1, 1]$ 2) $x \in R$
3) $x - (2n\pi + 1)\frac{\pi}{2}, n \in Z$ 4) $x = n\pi, n \in Z$

13. If $0 < x < 1$ then $\tan^{-1}\left(\frac{\sqrt{1-x^2}}{1+x}\right)$ is equal to

- 1) $\frac{1}{2}\cos^{-1}x$ 2) $\cos^{-1}\sqrt{\frac{1+x}{2}}$
3) $\sin^{-1}\sqrt{\frac{1-x}{2}}$ 4) All the above

C.U.Q-KEY

- 01) 1 02) 2 03) 3 04) 3 05) 3
06) 3 07) 3 08) 1 09) 1 10) 1
11) 3 12) 4 13) 4

C.U.Q-HINTS

- $\cos^{-1} x = \tan^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right)$
- $\tan^{-1} x = \cos^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right)$
- Apply $2 \tan^{-1} x$ formula
- $\cot^{-1} x = \sin^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right)$ & $\tan^{-1} x = \cos^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right)$
- $\cot^{-1}(-x) = \pi - \cot^{-1} x$
- Put $x = \tan A, y = \tan B, z = \tan C$
- Put $x = \tan A; y = \tan B$
- $2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta = \cos 2\theta$
- Put $x = \frac{1}{\sqrt{3}} = y = z$
- Apply $\cos^{-1} x + \cos^{-1} y$ formula
- Standard $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$
- $\operatorname{Cosec}^{-1}(\cos x)$ is defined only if $\cos x = \pm 1$
 $\Rightarrow x = n\pi; n \in \mathbb{Z}$
- Put $x = \frac{1}{2}, \text{G.E} = \frac{\pi}{6}$



LEVEL - I (C.W)

- The domain of $\sin^{-1} x + \cos^{-1} x$ is
 1) $(-\pi, \pi)$ 2) $[-1, 1]$ 3) $(0, 2\pi)$ 4) $(-\infty, \infty)$
- The domain of $\log_e \sin^{-1}(x)$ is
 1) $(0, 1]$ 2) $(0, 2]$ 3) $(0, \infty)$ 4) $(-\infty, 0]$
- The range of $\tan^{-1} x$ is
 1) $\mathbb{R}$ 2) $(0, \pi)$ 3) $[0, \pi]$ 4) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- The principal value of $\cos^{-1} \left(\cos \frac{7\pi}{6} \right)$ is
 1) $\frac{7\pi}{6}$ 2) $\frac{5\pi}{3}$ 3) $\frac{5\pi}{6}$ 4) $\frac{13\pi}{6}$
- $\sin^{-1} \left(\frac{\sqrt{3}}{2} \right) - \tan^{-1}(-\sqrt{3})$ is
 1) $-\frac{\pi}{3}$ 2) $\frac{2\pi}{3}$ 3) $\frac{\pi}{6}$ 4) 0

- $\cot^{-1}(2 + \sqrt{3}) =$
 1) $\frac{\pi}{12}$ 2) $\frac{\pi}{15}$ 3) $\frac{\pi}{5}$ 4) $\frac{3\pi}{10}$
- $\cot \left[\sin^{-1} \sqrt{\frac{13}{17}} \right] - \sin \left[\tan^{-1} \frac{2}{3} \right] =$
 1) $-\frac{2}{\sqrt{13}}$ 2) 0 3) $\frac{2}{\sqrt{13}}$ 4) $\frac{2}{3\sqrt{13}}$
- $\sec^2(\cot^{-1} \frac{1}{2}) + \operatorname{cosec}^2(\tan^{-1} \frac{1}{3}) =$
 1) 5 2) 10 3) 15 4) 50
- Find the value of $\sin \left(\sin^{-1} \frac{4}{5} + \sin^{-1} \frac{7}{25} \right)$
 1) $\frac{119}{125}$ 2) $\frac{117}{125}$ 3) $\frac{118}{125}$ 4) $\frac{113}{125}$
- $\cos(2\cos^{-1}(7/25)) =$
 1) $\frac{527}{625}$ 2) $-\frac{527}{625}$ 3) $\pi - \frac{527}{625}$ 4) $\frac{24}{25}$
- If $x + \frac{1}{x} = 2$, the principle value of $\sin^{-1} x$ is
 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) π 4) $\frac{3\pi}{2}$
- The numerical value of $\tan \left(2 \tan^{-1} \frac{1}{5} - \frac{\pi}{4} \right)$ is
 1) 1 2) 0 3) $\frac{7}{17}$ 4) $-\frac{7}{17}$
- $2 \tan^{-1}(1/2) + \sin^{-1}(3/5) =$
 1) $\tan^{-1} \left(\frac{12}{25} \right)$ 2) $\frac{\pi}{4}$
 3) $\frac{\pi}{2}$ 4) $\tan^{-1} \left(\frac{25}{12} \right)$
- $\tan^{-1}(2) + \tan^{-1}(3) =$
 1) $-\frac{\pi}{4}$ 2) $\frac{\pi}{4}$ 3) $\frac{3\pi}{4}$ 4) $\frac{5\pi}{4}$
- $\tan^{-1} \left(\frac{m}{n} \right) - \tan^{-1} \left(\frac{m-n}{m+n} \right) =$
 1) $\frac{\pi}{2}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{4}$ 4) $\frac{3\pi}{4}$

16. $2 \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7} =$
 1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{3\pi}{4}$
17. $\sec [\tan^{-1} 5 + \tan^{-1} \frac{1}{5} - \tan^{-1} \frac{3}{4}] =$
 1) $3/5$ 2) $5/3$ 3) $4/5$ 4) $\sqrt{2}$
18. $4 \tan^{-1} \frac{1}{5} - \tan^{-1} \frac{1}{70} + \tan^{-1} \frac{1}{99} =$
 1) π 2) $\pi/2$ 3) $\pi/4$ 4) $3\pi/4$
19. $\tan^{-1} \left(\frac{5}{12} \right) + \sin^{-1} \left(\frac{24}{25} \right) = \cos^{-1}(x) \Rightarrow x =$
 1) $\frac{-31}{325}$ 2) $\frac{-33}{325}$ 3) $\frac{-36}{325}$ 4) $\frac{-39}{325}$
20. If $\sin^{-1} \left(\frac{12}{13} \right) + \sec^{-1} \left(\frac{13}{x} \right) = \frac{\pi}{2}$ then $x =$
 1) 12 2) 13 3) 11 4) 5
21. If $\sin^{-1} \left(\frac{3}{x} \right) + \sin^{-1} \left(\frac{4}{x} \right) = \frac{\pi}{2}$ then $x =$ (EAM-2008)
 1) 6 2) 7 3) 8 4) 5
22. $\tan^{-1} \left(\frac{a}{x} \right) + \tan^{-1} \left(\frac{b}{x} \right) = \frac{\pi}{2}$ then $x =$
 1) ab 2) $\sqrt{ab}$ 3) $\sqrt{a^2 + b^2}$ 4) $a^2 + b^2$
23. $\tan^{-1} \left(x + \sqrt{1+x^2} \right) =$
 1) $\frac{\pi}{4} - \frac{1}{2} \tan^{-1} x$ 2) $\frac{1}{2} \tan^{-1} x$
 3) $\frac{\pi}{2} - \frac{1}{2} \tan^{-1} x$ 4) $\frac{\pi}{4} + \frac{1}{2} \tan^{-1} x$
24. If $\tan^{-1} \left(\frac{3a^2x - x^3}{a^3 - 3ax^2} \right) = k \tan^{-1} \left(\frac{x}{a} \right)$ then $k =$
 1) 2 2) 3 3) -2 4) 4
25. If $\tan^{-1} \left(\frac{2x}{x^2 - 1} \right) + \cos^{-1} \frac{x^2 - 1}{x^2 + 1} = \frac{2\pi}{3}$ then $x =$
 1) $2 - \sqrt{3}$ 2) $\sqrt{3} - \sqrt{2}$ 3) $2 + \sqrt{3}$ 4) $+\sqrt{2}$
26. If $\tan^{-1} \left(\frac{x-1}{x-2} \right) + \tan^{-1} \left(\frac{x+1}{x+2} \right) = \frac{\pi}{4}$ then $x =$
 1) $\frac{1}{\sqrt{2}}$ 2) $\pm \frac{1}{\sqrt{2}}$ 3) $\pm \frac{1}{\sqrt{3}}$ 4) $\frac{1}{\sqrt{3}}$

27. If $\cos^{-1} x = \cot^{-1} (4/3) + \tan^{-1} (1/7)$ then $x =$

- 1) $1/2$ 2) $\frac{\sqrt{3}}{2}$ 3) $\frac{1}{\sqrt{2}}$ 4) $\frac{3}{5}$

28. The trigonometric equation $\sin^{-1} x = 2 \sin^{-1} a$ has a solution for (AIEEE-2003)

- 1) $|a| \leq \frac{1}{\sqrt{2}}$ 2) all real values of a

- 3) $|a| < \frac{1}{2}$ 4) $|a| \geq \frac{1}{\sqrt{2}}$

29. If $n \in \mathbb{N}$, $\sum_{k=1}^n \sin^{-1}(x_k) = \frac{n\pi}{2}$ then $\sum_{k=1}^n x_k =$

- 1) n 2) k 3) $\frac{k(k+1)}{2}$ 4) $\frac{n(n+1)}{2}$

30. If $\sin^{-1} \left(x - \frac{x^2}{2} + \frac{x^3}{4} - \dots \right) + \cos^{-1} \left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \dots \right)$

$= \frac{\pi}{2}$ for $0 < |x| < \sqrt{2}$ then $x =$

- 1) $1/2$ 2) 1 3) $-1/2$ 4) -1

31. The value of $\sin^{-1}(\sin 10)$ is

- 1) 10 2) $10 - 3\pi$ 3) $3\pi - 10$ 4) -10

32. The greatest of $\tan^{-1} 1, \sin^{-1} 1, \sin 1, \cos 1$ is

- 1) $\sin 1$ 2) $\cos 1$ 3) $\tan^{-1} 1$ 4) $\sin^{-1} 1$

33. If $\tan^{-1} x, \tan^{-1} y, \tan^{-1} z$ are in A.P. then $\frac{2y}{1-y^2} =$

- 1) $\frac{x-z}{1+xz}$ 2) $\frac{x+z}{1-xz}$ 3) $x+z$ 4) xz

LEVEL - I (C.W)-KEY

- 01) 2 02) 1 03) 4 04) 3 05) 2 06) 1
 07) 2 08) 3 09) 2 10) 2 11) 2 12) 4
 13) 3 14) 3 15) 3 16) 3 17) 2 18) 3
 19) 3 20) 1 21) 4 22) 2 23) 4 24) 2
 25) 1 26) 2 27) 3 28) 1 29) 1 30) 2
 31) 3 32) 4 33) 2

LEVEL - I (C.W)-HINTS

- Domain $= [-1, 1]$
- Domain of $\log_e x$ is $x > 0$
- Range of $\tan^{-1} x = \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$

INVERSE TRIGONOMETRIC FUNCTIONS

MATHS BY Er. RG SIR

4. $\cos^{-1}[\cos(\frac{7\pi}{6})] = \cos^{-1}[\cos(2\pi - \frac{5\pi}{6})] = \frac{5\pi}{6}$
5. $\sin^{-1}(\frac{\sqrt{3}}{2}) - \tan^{-1}(-\sqrt{3}) = \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$
6. $\cot 15^\circ = 2 + \sqrt{3}$
7. $\sin^{-1}\sqrt{\frac{13}{17}} = \cot^{-1}\frac{2}{\sqrt{13}}, \tan^{-1}\frac{2}{3} = \sin^{-1}\frac{2}{\sqrt{13}}$
8. $\sec^2(\tan^{-1} 2) + \operatorname{cosec}^2(\cot^{-1} 3) =$
 $1 + \tan^2(\tan^{-1} 2) + 1 + \cot^2(\cot^{-1} 3)$
9. Apply $\sin^{-1}x + \sin^{-1}y = \sin^{-1}[x\sqrt{1-y^2} + y\sqrt{1-x^2}]$
 formula where $x = \frac{4}{5}, y = \frac{7}{25}$
10. Apply $2\cos^{-1}x = \cos^{-1}(2x^2 - 1)$ formula
 where $x = \frac{7}{25}$
11. Put $X=1$
12. Apply $2\tan^{-1}x = \tan^{-1}(\frac{2x}{1-x^2})$ formula
 where $X = 1/5$
13. $2\tan^{-1}(\frac{1}{2}) = \cos^{-1}\left(\frac{1-\frac{1}{4}}{1+\frac{1}{4}}\right) = \cos^{-1}(\frac{3}{5})$
14. $\tan^{-1}x + \tan^{-1}y = \pi + \tan^{-1}(\frac{x+y}{1-xy}),$
 $x>0, y>0 \quad xy>1$ formula
15. Apply $\tan^{-1}x - \tan^{-1}y = \tan^{-1}(\frac{x-y}{1+xy})$, formula
16. Apply $2\tan^{-1}x$ formula after
 $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$
17. $\tan^{-1}5 + \tan^{-1}\frac{1}{5} = \frac{\pi}{2}$ and $\tan^{-1}\frac{3}{4} = \operatorname{cosec}^{-1}\frac{5}{3}$
18. $4\tan^{-1}\frac{1}{5} = \tan^{-1}\frac{120}{119}$ Apply
 $\tan^{-1}x - \tan^{-1}y$ and $\tan^{-1}x + \tan^{-1}y$

19. $\sin^{-1}(\frac{24}{25}) = \tan^{-1}(\frac{24}{7})$ and apply
 $\tan^{-1}x + \tan^{-1}y = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right); xy > 1$
20. $\sec^{-1}x = \cos^{-1}\frac{1}{x}$ and $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$
21. Apply $\sin^{-1}\frac{a}{x} + \sin^{-1}\frac{b}{x} = \frac{\pi}{2}$ then $x = \sqrt{a^2 + b^2}$
22. Apply $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2} \Rightarrow \frac{a}{x} = \frac{x}{b} \Rightarrow x = \sqrt{ab}$
23. Put $x = \tan\theta$
24. Put $\frac{x}{a} = \tan\theta$
25. Put $x = \tan\theta$
26. Apply $\tan^{-1}x + \tan^{-1}y$ formula
27. $\cos^{-1}x = \tan^{-1}\frac{3}{4} + \tan^{-1}(\frac{1}{7})$ Apply
 $\tan^{-1}\left(\frac{p}{q}\right) + \tan^{-1}\left(\frac{q-p}{q+p}\right) = \frac{\pi}{4}$
 $\Rightarrow \cos^{-1}x = \frac{\pi}{4} \Rightarrow x = \frac{1}{\sqrt{2}}$
28. The range of $\sin^{-1}x = [-\frac{\pi}{2}, \frac{\pi}{2}]$
 $\therefore -\frac{\pi}{2} \leq \sin^{-1}x \leq \frac{\pi}{2}$
 $\Rightarrow -\frac{\pi}{2} \leq 2\sin^{-1}a \leq \frac{\pi}{2} \Rightarrow -\frac{\pi}{4} \leq \sin^{-1}a \leq \frac{\pi}{4}$
 $\Rightarrow -\frac{1}{\sqrt{2}} \leq a \leq \frac{1}{\sqrt{2}} \Rightarrow |a| \leq \frac{1}{\sqrt{2}}$
29. Put $x_1 = x_2 = x_3 = \dots = 1$
 $\sum_{k=1}^n x_k = 1 + 1 + 1 + \dots + 1 (n \text{ time}) = n$
30. Put $x = x^2 \Rightarrow x(x-1) = 0 \Rightarrow x = 0$ (or) 1
31. radian = $\frac{7\pi}{22} = 57^\circ 17' 44.8''$ and $\sin^{-1}$ lies
 between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$

32. 1 radian = $57^0 17' 44.8''$ $\sin^{-1} 1 = \frac{\pi}{2}$ is the greatest

33. $2\tan^{-1} y = \tan^{-1} x + \tan^{-1} z$

$$\tan^{-1}\left(\frac{2y}{1-y^2}\right) = \tan^{-1}\left(\frac{x+z}{1-xz}\right)$$



LEVEL - I (H.W)

- The domain of $\sin^{-1} \frac{2x+1}{3}$ is
1. $(-2, 1]$ 2. $[-2, 1]$ 3. $\mathbb{R}$ 4. $[-1, 1]$
- The domain of $\cos^{-1} \sqrt{2x}$ is
1) $[-1, 1]$ 2) $[-1/2, 1/2]$
3) $[0, 1/2]$ 4) $(1, 1/2)$
- The range of $\cot^{-1} x$ is
1) $(0, \pi)$ 2) $[0, \pi]$ 3) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ 4) $\mathbb{R}$
- The principal value of $\sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right)$ is
1) $\frac{2\pi}{3}$ 2) $\frac{\pi}{3}$ 3) $\frac{-\pi}{3}$ 4) $\frac{-2\pi}{3}$
- The principal value of $\sin^{-1}\left(\tan\left(\frac{-5\pi}{4}\right)\right)$ is
1) $\frac{\pi}{4}$ 2) $\frac{-\pi}{4}$ 3) $\frac{\pi}{2}$ 4) $\frac{-\pi}{2}$
- $\sec^{-1}\left(\frac{2\sqrt{2}}{1+\sqrt{3}}\right)$
1) $\frac{\pi}{12}$ 2) $\frac{\pi}{3}$ 3) $\frac{3\pi}{4}$ 4) $\frac{\pi}{6}$
- The value of $\sin(2\tan^{-1} \frac{1}{3}) + \cos(\tan^{-1} 2\sqrt{2})$ is
1) $\frac{12}{13}$ 2) $\frac{13}{14}$ 3) $\frac{14}{15}$ 4) $\frac{16}{15}$
- $\sec^2(\tan^{-1}(2)) + \operatorname{cosec}^2(\cot^{-1}(2)) =$
1) 5 2) 10 3) 15 4) 50
- $\cos^{-1}\left(\frac{4}{5}\right) + \cos^{-1}\left(\frac{63}{65}\right) =$
1) $\cos^{-1}\left(\frac{204}{325}\right)$ 2) $\cos^{-1}\left(\frac{300}{325}\right)$
3) $\cos^{-1}\left(\frac{201}{300}\right)$ 4) $\sin^{-1}\left(\frac{204}{325}\right)$

- $\sin(1/2 \cot^{-1}(-3/4)) =$
1) $1/\sqrt{5}$ 2) $2/\sqrt{5}$ 3) $-2/\sqrt{5}$ 4) $-1/\sqrt{5}$
- If 'x' is a negative real number then the value of $\cos^{-1} x + \sec^{-1} x$ is
1) rational number 2) irrational number
3) integer 4) imaginary
- $\tan(2 \tan^{-1}(\frac{\sqrt{5}-1}{2})) =$
1) 4 2) 3 3) $1/2$ 4) 2
- $\sin^{-1} \frac{4}{5} + 2 \tan^{-1} \frac{1}{3} =$ (Eam-2005)
1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{3\pi}{4}$
- $\tan^{-1}(\frac{1}{2}) + \tan^{-1}(\frac{1}{3}) =$ (Eam-1999)
1) $\frac{\pi}{2}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{6}$
- $\tan^{-1}(\frac{3}{2}) - \tan^{-1}(\frac{1}{5}) =$
1) $\frac{-\pi}{4}$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{3\pi}{4}$
- $4 \tan^{-1} \frac{1}{5} - \tan^{-1} \frac{1}{239} =$
1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{3\pi}{4}$
- $\cot(\operatorname{cosec}^{-1} \frac{5}{3} + \tan^{-1} \frac{2}{3}) =$
1) $\frac{6}{17}$ 2) $\frac{3}{17}$ 3) $\frac{4}{17}$ 4) $\frac{5}{17}$
- $2 \tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{7} + 2 \tan^{-1} \frac{1}{8} =$
1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{3\pi}{4}$
- If $\sin^{-1} x - \cos^{-1} x = \frac{\pi}{6}$ then $x =$ (Eam-2002)
1) $1/2$ 2) $\frac{\sqrt{3}}{2}$ 3) $-1/2$ 4) $-\frac{\sqrt{3}}{2}$

20. $\cos^{-1}(\sqrt{3}x) + \cos^{-1}x = \frac{\pi}{2}$ then $x =$

- 1) 0 2) $\frac{1}{\sqrt{2}}$ 3) $\frac{1}{2}$ 4) 1

21. $\sin^{-1}\left(\frac{5}{x}\right) + \sin^{-1}\left(\frac{12}{x}\right) = \frac{\pi}{2}$, ($x > 0$) then $x =$

- 1) 12 2) 13 3) 14 4) 15

22. $\sin^{-1}\left(\frac{x}{5}\right) + \operatorname{cosec}^{-1}\left(\frac{5}{4}\right) = \frac{\pi}{2}$ then $x =$ (AIE-2007)

- 1) 1 2) 3 3) 4 4) 5

23. $\tan^{-1}\left(\frac{1 + \sin x}{\cos x}\right) =$

- 1) $\frac{\pi}{4} - \frac{x}{2}$ 2) $\frac{\pi}{4} - x$ 3) $\frac{\pi}{4} + x$ 4) $\frac{\pi}{4} + \frac{x}{2}$

24. A value of $\tan^{-1}\left\{\sin\left(\cos^{-1}\sqrt{\frac{2}{3}}\right)\right\}$ is

- 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{6}$

25. $\tan\left[\frac{1}{2}\sin^{-1}\left(\frac{2a}{1+a^2}\right) + \frac{1}{2}\cos^{-1}\left(\frac{1-a^2}{1+a^2}\right)\right] =$

- 1) $\frac{2a}{1+a^2}$ 2) $\frac{2a}{1-a^2}$ 3) $\frac{a}{1+a^2}$ 4) $\frac{a}{1-a^2}$

26. If $3\tan^{-1}(2 - \sqrt{3}) - \tan^{-1}(x) = \tan^{-1}(1/3)$ then $x =$

- 1) 1/2 2) 2 3) 3 4) 1/3

27. If $\cos^{-1}\left(\frac{8}{17}\right) - \cos^{-1}\left(\frac{5}{13}\right) = \cos^{-1}x$ then $x =$

- 1) $\frac{140}{221}$ 2) $\frac{220}{221}$ 3) $\frac{7}{11}$ 4) $\frac{221}{220}$

28. The equation $2\cos^{-1}x = \cos^{-1}(2x^2 - 1)$ is satisfied by

- 1) $-1 \leq x \leq 1$ 2) $0 \leq x \leq 1$
3) $x \geq 1$ 4) $x \leq 1$

29. If $\sum_{r=1}^n \cos^{-1}x_r = 0$, then $\sum_{r=1}^n x_r$ equals to

- 1) 0 2) n 3) $\frac{n(n+1)}{2}$ 4) $\frac{n}{2}$

30. $\sin^{-1}\left[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}\right] =$

- 1) $\sin^{-1}x + \sin^{-1}\sqrt{x}$ 2) $\sin^{-1}x - \sin^{-1}\sqrt{x}$
3) $\sin^{-1}\sqrt{x} - \sin^{-1}x$ 4) 0

31. If $\theta \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ then $\sin^{-1}(\sin \theta) =$

- 1) θ 2) $\pi - \theta$ 3) $2\pi - \theta$ 4) $-\pi + \theta$

32. $\cos^{-1}\left(\frac{a-b}{a+b}\right) =$

- 1) $2\tan^{-1}b$ 2) $2\tan^{-1}\sqrt{\frac{b}{a}}$
3) $2\tan^{-1}\sqrt{\frac{a}{b}}$ 4) $2\tan^{-1}\left(\frac{ab}{a-b}\right)$

33. If $\theta = \tan^{-1}a$, $\phi = \tan^{-1}b$ and $ab = -1$, then $\theta - \phi$ is equal to :

- 1) 0 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{2}$ 4) π

LEVEL - I (H.W)-KEY

- 01) 2 02) 3 03) 1 04) 2 05) 4 06) 1
07) 3 08) 2 09) 1 10) 2 11) 2 12) 4
13) 2 14) 2 15) 3 16) 3 17) 1 18) 3
19) 2 20) 3 21) 2 22) 2 23) 4 24) 4
25) 2 26) 1 27) 2 28) 2 29) 2 30) 2
31) 2 32) 2 33) 3

LEVEL - I (H.W)-HINTS

- $\sin^{-1}x$ domain $[-1, 1] \Rightarrow -1 \leq \frac{2x+1}{3} \leq 1$
- $0 \leq 2x \leq 1$
- Range of $\cot^{-1}x$ is $(0, \pi)$
- $\sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right) = \sin^{-1}\left[\sin\left(\pi - \frac{\pi}{3}\right)\right]$
- $\sin^{-1}\left[\tan\left(\frac{-5\pi}{4}\right)\right] = \sin^{-1}(-1)$
- $\frac{2\sqrt{2}}{1+\sqrt{3}} = \sec \frac{\pi}{12}$
- Apply $2\tan^{-1}x = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ and $\tan^{-1}x = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$ formula

8. $\sec^2(\tan^{-1}2) + \operatorname{cosec}^2(\cot^{-1}2)$
 $= 1 + \tan^2(\tan^{-1}2) + 1 + \cot^2(\cot^{-1}2)$
9. A p p l y
 $\cos^{-1}x + \cos^{-1}y = \cos^{-1}(xy - \sqrt{1-x^2}\sqrt{1-y^2})$
 formula
10. Put $\cot^{-1}(\frac{-3}{4}) = x$
11. $x = -1$ only satisfied
 $\therefore \cos^{-1}x + \sec^{-1}x = 2\pi = \text{Irrational number}$
12. Apply $2\tan^{-1}x$ formula
13. Apply $2\tan^{-1}x$ formula
 $\therefore \sin^{-1}\frac{4}{5} + 2\tan^{-1}\frac{1}{3} = \sin^{-1}\frac{4}{5} + \cos^{-1}\frac{4}{5} = \frac{\pi}{2}$
14. Apply $\tan^{-1}(\frac{p}{q}) + \tan^{-1}(\frac{q-p}{q+p}) = \frac{\pi}{4}$
15. Apply $\tan^{-1}x - \tan^{-1}y$
16. $4\tan^{-1}(\frac{1}{5}) = \tan^{-1}(\frac{120}{19})$ then apply
 $\tan^{-1}x - \tan^{-1}y$ formula
17. $\cot^{-1}(\operatorname{cosec}^{-1}\frac{5}{3} + \tan^{-1}\frac{2}{3})$
 $= \cot(\sin^{-1}(\frac{3}{5}) + \tan^{-1}(\frac{2}{3}))$
 Apply $\sin^{-1}x = \tan^{-1}\frac{x}{\sqrt{1-x^2}}$ and
 $\Rightarrow \sin^{-1}\frac{3}{5} = \tan^{-1}\frac{3}{4}$ apply $\tan^{-1}x + \tan^{-1}y$
 formula
18. Apply $2\tan^{-1}x = \tan^{-1}(\frac{2x}{1-x^2})$ and
 $\tan^{-1}x + \tan^{-1}y$ formula
19. Put $x = \frac{\sqrt{3}}{2}$
20. Put $x = \frac{1}{2}$

21. Apply $\sin^{-1}(\frac{a}{x}) + \sin^{-1}(\frac{b}{x}) = \frac{\pi}{2}$ then
 $x = \sqrt{a^2 + b^2}$ formula
22. $\sin^{-1}\frac{x}{5} + \sin^{-1}(\frac{4}{5}) = \frac{\pi}{2}$
 Apply $\sin^{-1}\frac{a}{x} + \sin^{-1}\frac{b}{x} = \frac{\pi}{2}$ then
 $x = \sqrt{a^2 + b^2} \Rightarrow 5 = \sqrt{x^2 + 4^2} \Rightarrow x = 3$
23. $\frac{1 + \sin x}{\cos x} = \tan(\frac{\pi}{4} + \frac{x}{2})$
24. $\tan^{-1}[\sin(\sin^{-1}\sqrt{\frac{1}{3}})] = \tan^{-1}(\frac{1}{\sqrt{3}}) = \frac{\pi}{6}$
25. Put $a = \tan\theta$
26. $2 - \sqrt{3} = \tan 15^\circ$ then $\tan^{-1}x = \frac{\pi}{4} - \tan^{-1}\frac{1}{3}$
27. Apply $\cos^{-1}x - \cos^{-1}y = \cos^{-1}(xy + \sqrt{1-x^2}\sqrt{1-y^2})$
28. Given equation satisfied for $0 \leq x \leq 1$
29. $\cos^{-1}x_1 = \cos^{-1}x_2 = \dots = \cos^{-1}x_n = 0$
 $\Rightarrow x_1 = x_2 = \dots = x_n = +1$
 $\therefore \sum_{r=1}^n x_r = n$
30. $\sin^{-1}[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}] = \sin^{-1}[x\sqrt{1-(\sqrt{x})^2} - \sqrt{x}\sqrt{1-x^2}]$
 $= \sin^{-1}x - \sin^{-1}\sqrt{x}$
31. $\sin^{-1}(\sin\theta) = \sin^{-1}(\sin(\pi - \theta)) = \pi - \theta$
32. $\cos^{-1}\left(\frac{1 - \frac{b}{a}}{1 + \frac{b}{a}}\right) = \cos^{-1}\left(\frac{1 - \left(\sqrt{\frac{b}{a}}\right)^2}{1 + \left(\sqrt{\frac{b}{a}}\right)^2}\right) = 2\tan^{-1}\sqrt{\frac{b}{a}}$
33. $\tan(\theta - \Phi) = \frac{\tan\theta - \tan\Phi}{1 + \tan\theta\tan\Phi} = \frac{a-b}{1-1} = \infty$
 $\theta - \Phi = \frac{\pi}{2}$



1. Range of $\sin^{-1}x + \cos^{-1}x + \tan^{-1}x$ is

- 1) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ 2) $(0, \pi]$ 3) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ 4) $[0, \pi]$

2. Range of $f(x) = \sin^{-1}x + \tan^{-1}x + \sec^{-1}x$ is

- 1) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ 2) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$

- 3) $\left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$ 4) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

3. The domain of $\sin^{-1}[\log_2(x^2/2)]$ is

- 1) $[-2, -1]$ 2) $[1, 2]$
3) $[-2, -1] \cup [1, 2]$ 4) $[-2, 0]$

4. Arrange the following functions in the increasing order of their domains.

$$f(x) = \sin^{-1}x; g(x) = \cos^{-1}\sqrt{2x}; h(x) = \sin^{-1}2x; k(x) = \tan^{-1}(5x)$$

- 1) $f(x), g(x), h(x), k(x)$ 2) $g(x), h(x), k(x), f(x)$
3) $g(x), h(x), f(x), k(x)$ 4) $g(x), f(x), h(x), k(x)$

5. The value of $\cos(2\cos^{-1}0.8)$ is

- 1) 0.48 2) 0.96 3) 0.6 4) 0.28

6. The value of the expression

$$\sin^{-1}\left(\sin\frac{22\pi}{7}\right) + \cos^{-1}\left(\cos\frac{5\pi}{3}\right) + \tan^{-1}\left(\tan\frac{5\pi}{7}\right) + \sin^{-1}(\cos 2)$$

is

- 1) $\frac{17\pi}{42} - 2$ 2) -2 3) $-\frac{\pi}{21} - 2$ 4) $\frac{\pi}{21}$

7. $\cos\left[2\sin^{-1}\sqrt{\frac{1-x}{2}}\right] =$

- 1) x 2) $\frac{1}{x}$ 3) $2x$ 4) $3x$

8. If $\sin^{-1}(x) - \cos^{-1}(x) = \sin^{-1}(x-1)$ then $x =$
[EAM-2004]

- 1) $0, \frac{1}{2}$ 2) $-1, \frac{1}{2}$ 3) $1, -\frac{1}{2}$ 4) $\frac{1}{2}, 1$

9. If $\sin^{-1}(6x) + \sin^{-1}(6\sqrt{3}x) = -\frac{\pi}{2}$ then the value of x is

- 1) $\frac{1}{12}$ 2) $-\frac{1}{12}$ 3) $-\frac{1}{4\sqrt{3}}$ 4) $\frac{1}{4\sqrt{3}}$

10. If $\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$ then $x =$

- 1) $1/2$ 2) $1/4$ 3) $1/6$ 4) 6

11. If $\cot^{-1}\left(\frac{1-x^2}{2x}\right) + \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \frac{2\pi}{3}$; $x > 0$; $x \neq 1$ then $x =$

- 1) $\frac{1}{\sqrt{2}}$ 2) $\pm\frac{1}{\sqrt{2}}$ 3) $\pm\frac{1}{\sqrt{3}}$ 4) $\frac{1}{\sqrt{3}}$

12. If $\tan^{-1}\frac{1}{1+2x} + \tan^{-1}\frac{1}{4x+1} = \tan^{-1}\frac{2}{x^2}$ then $x =$

- 1) 1 2) 0 3) -3 4) $\frac{2}{3}$

13. $\tan\left[\tan^{-1}\left(\frac{1}{a+b}\right) + \tan^{-1}\left(\frac{b}{a^2+ab+1}\right)\right] =$

- 1) a 2) $1/a$ 3) b 4) $1/b$

14. In a $\triangle ABC$, If C is a right angle then

$$\tan^{-1}\left(\frac{a}{b+c}\right) + \tan^{-1}\left(\frac{b}{c+a}\right) =$$

- 1) $\frac{\pi}{3}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{6}$ 4) $\frac{5\pi}{2}$

15. If $\tan^{-1}\left(\frac{x+1}{x-1}\right) + \tan^{-1}\left(\frac{x-1}{x}\right) = \pi + \tan^{-1}(-7)$ then $x =$

- 1) 2 2) -2 3) 1 4) no solution

16. The number of real solutions of

$$\tan^{-1}(\sqrt{x(x+1)}) + \sin^{-1}\sqrt{(x^2+x+1)} = \frac{\pi}{2}$$

- 1) 0 2) 1 3) 2 4) ∞

17. If $x = \tan^{-1}(1) + \cos^{-1}(-\frac{1}{2}) + \sin^{-1}(-\frac{1}{2})$ and $y = \cos\left[\frac{1}{2}\cos^{-1}(1/8)\right]$ then

- 1) $x = 2\pi$ 2) $y = 3\pi$ 3) $x = \pi$ 4) $y = \pi$

18. Let a, b, c be a positive real numbers

$$\theta = \tan^{-1}\sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1}\sqrt{\frac{b(a+b+c)}{ca}} +$$

$$\tan^{-1}\sqrt{\frac{c(a+b+c)}{ab}}$$

- 1) 0 2) 3π 3) 1 4) 4π

19. If x, y and z are in A.P and $\tan^{-1}x, \tan^{-1}y$ and $\tan^{-1}z$ are also in A.P then (Aie-2013)

- 1) $x = y = z$ 2) $2x = 3y = 6z$
3) $6x = 3y = 2z$ 4) $6x = 4y = 3z$

20. If $\left[\sin^{-1} \left(\cos^{-1} \left(\sin^{-1} \left(\tan^{-1} x \right) \right) \right) \right] = 1$, where $[\bullet]$ denotes the greatest integer function, then $x \in$

- 1) $[\tan \sin \cos 1, \tan \sin \cos \sin 1]$
 2) $(\tan \sin \cos 1, \tan \sin \cos \sin 1)$
 3) $[-1, 1]$ 4) $[\sin \cos \tan 1, \sin \cos \sin \tan 1]$

21. If $x(3 - x) \geq 2$ then $\sin^{-1}(x) + \sin^{-1}(x^2) + \dots + \sin^{-1}(x^{10}) =$

- 1) $\frac{\pi}{2}$ 2) 2π 3) 5π 4) 10π

22. If $\cos^{-1} x + \cos^{-1} y = \frac{\pi}{2}$ and $\tan^{-1} x - \tan^{-1} y = 0$ then $x^2 + xy + y^2 =$

- 1) 0 2) $-\frac{1}{2}$ 3) $\frac{1}{2}$ 4) $\frac{3}{2}$

23. If $0 < x < 1$, then $\sqrt{1+x^2} \left[\left\{ x \cos(\cot^{-1} x) + \sin(\cot^{-1} x) \right\}^2 - 1 \right] =$

- 1) $\frac{x}{\sqrt{1+x^2}}$ 2) x 3) $x\sqrt{1+x^2}$ 4) $\sqrt{1+x^2}$

24. The value of 'a' for which $ax^2 + \sin^{-1}(x^2 - 2x + 2) + \cos^{-1}(x^2 - 2x + 2) = 0$ has a real solution, is

- 1) $\frac{\pi}{2}$ 2) $-\frac{\pi}{2}$ 3) $\frac{2}{\pi}$ 4) $-\frac{2}{\pi}$

25. If minimum value of $(\sin^{-1} x)^2 + (\cos^{-1} x)^2$ is $\frac{\pi^2}{k}$, then value of k is

- 1) 4 2) 6 3) 8 4) 10

26. If $(\tan^{-1} x)^2 + (\cot^{-1} x)^2 = \frac{5\pi^2}{8}$, then $x =$

- [Eam-2011]
 1) -1 2) 1 3) 0 4) 2

27. If $6\sin^{-1}(x^2 - 6x + 12) = 2\pi$, then the value of x , is

- 1) 1 2) 2 3) 3 4) does not exist

28. If $\cos^{-1} x > \sin^{-1} x$, then x belongs to the interval

- 1) $(-\infty, 0)$ 2) $(-1, 0)$
 3) $[0, \frac{1}{\sqrt{2}})$ 4) $[-1, \frac{1}{\sqrt{2}})$

29. The least integral value of x for which $\tan^{-1} x > \cot^{-1} x$ is

- 1) 1 2) 2 3) 3 4) 4

30. The value of $\sin^{-1}(\sin 12) + \sin^{-1}(\cos 12) =$

- 1) 0 2) $24 - 2\pi$ 3) $4\pi - 24$ 4) 8π

31. If $\cos^{-1} x - \cos^{-1}(\frac{y}{2}) = \alpha$ then $4x^2 - 4xy \cos \alpha + y^2 =$ [AIEEE-2005]

- 1) $-4\sin^2 \alpha$ 2) $4\sin^2 \alpha$
 3) 4 4) $2\sin 2\alpha$

32. If $\log_2 x \geq 0$ then $\log_{\frac{1}{\pi}} \left\{ \sin^{-1} \frac{2x}{1+x^2} + 2\tan^{-1} x \right\} =$

- 1) $\log_{\frac{1}{\pi}}(4\tan^{-1} x)$ 2) 0 3) -1 4) -2

33. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$ then $\sum \left(\frac{x^{201} + y^{201}}{x^{603} + y^{603}} \right) \left(\frac{x^{402} + y^{402}}{x^{804} + y^{804}} \right) =$

- 1) 0 2) 1 3) 2 4) 3

34. The ascending order of minimum values of the function $P: \sin^{-1} x - \cos^{-1} x$

$Q: \tan^{-1} x + \cot^{-1} x$, $R: \sec^{-1} x - \operatorname{cosec}^{-1} x$

- 1) P, Q, R 2) P, R, Q
 3) Q, P, R 4) Q, R, P

35. $\tan^{-1} \left(\frac{1}{1 + (1)(2)} \right) + \tan^{-1} \left(\frac{1}{1 + (2)(3)} \right) +$

$\dots + \tan^{-1} \left(\frac{1}{1 + (n-1)(n)} \right) =$

- 1) 0 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{2}$ 4) $\tan^{-1} \left(\frac{n-1}{n+1} \right)$

LEVEL - II (C.W)-KEY

- 01) 3 02) 3 03) 3 04) 3 05) 4
 06) 1 07) 1 08) 1 09) 2 10) 3
 11) 4 12) 2 13) 2 14) 2 15) 1
 16) 3 17) 3 18) 1 19) 1 20) 1
 21) 3 22) 4 23) 3 24) 2 25) 3
 26) 1 27) 4 28) 4 29) 2 30) 1
 31) 2 32) 3 33) 4 34) 2 35) 4

LEVEL - II (C.W)-HINTS

1. Domain of $\sin^{-1}x + \cos^{-1}x + \tan^{-1}x$ is $[-1, 1]$

$$\text{Range is } \left[\frac{\pi}{2} + \tan^{-1}(-1), \frac{\pi}{2} + \tan^{-1}(1)\right]$$

2. Common domain = $\{-1, 1\}$

$$\text{range} = \{f(-1), f(1)\}$$

3. $-1 \leq \log_2 \left(\frac{x^2}{2}\right) \leq 1 \Rightarrow 2^{-1} \leq \frac{x^2}{2} \leq 2 \Rightarrow 1 \leq x^2 \leq 4$

4. f: $\sin^{-1}(x)$ domain $[-1, 1]$

$$\text{g: } \cos^{-1} \sqrt{2x} \text{ domain } \left[0, \frac{1}{2}\right]$$

$$\text{h: } \sin^{-1} 2x \text{ domain } \left[-\frac{1}{2}, \frac{1}{2}\right]$$

$$\text{k: } \tan^{-1} 5x \text{ domain } \mathbb{R}$$

5. Apply $2\cos^{-1}x = \cos^{-1}(2x^2 - 1)$ formula

$$6. -\frac{\pi}{7} + \frac{\pi}{3} - \frac{2\pi}{7} + \left(\frac{\pi}{2} - 2\right) = \frac{17\pi}{42} - 2$$

$$7. \text{ Put } x = \cos\theta \Rightarrow \sqrt{\frac{1-x}{2}} = \sin\theta$$

8. Verify with option

9. Apply $\sin^{-1}x + \sin^{-1}y$ formula

10. Apply $\tan^{-1}x + \tan^{-1}y$ formula

11. Put $x = \tan\theta$

12. Apply $\tan^{-1}x + \tan^{-1}y$ formula

$$\begin{aligned} 13. \quad & \tan\left[\tan^{-1}\left(\frac{1}{a+b}\right) + \tan^{-1}\left(\frac{(a+b)-a}{1+(a+b)a}\right)\right] \\ &= \tan\left[\tan^{-1}\left(\frac{1}{a+b}\right) + \tan^{-1}(a+b) - \tan^{-1}a\right] \\ &= \tan\left[\frac{\pi}{2} - \tan^{-1}(a)\right] = \frac{1}{a} \end{aligned}$$

14. Applying $\tan^{-1}x + \tan^{-1}y$ formula and substitute $c^2 = a^2 + b^2$

15. By verification $x=2$ satisfied

16. $x(x+1) \geq 0$ and $x^2 + x + 1 \geq 0$

$$\text{But } \sin^{-1}(\sqrt{x}) \text{ domain } [0, 1]$$

$$\therefore x(x+1) = 0 \Rightarrow x = 0 \text{ or } -1$$

$$17. \quad x = \frac{3\pi}{4}, \quad y = \cos \frac{\alpha}{2} \text{ put}$$

$$\cos^{-1}\left(\frac{1}{8}\right) = \alpha, \quad y = \sqrt{\frac{1+\cos\alpha}{2}} = \frac{3}{4}$$

$$18. \text{ use } \tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \tan^{-1}\left(\frac{\sum x - \prod x}{1 - \sum xy}\right)$$

(or) Put $a=b=c=1$ verify option

$$19. \quad 2\tan^{-1}y = \tan^{-1}x + \tan^{-1}z \Rightarrow y^2 = xz$$

$\therefore x, y, z$ are in G.P as well as in A.P

$$x = y = z$$

$$20. \quad 1 \leq \sin^{-1}\cos^{-1}\sin^{-1}\tan^{-1}x \leq \frac{\pi}{2}$$

$$\Rightarrow \sin 1 \leq \cos^{-1}\sin^{-1}\tan^{-1}x \leq 1$$

$$\Rightarrow \cos \sin 1 \geq \sin^{-1}\tan^{-1}x \geq \cos 1$$

$$\Rightarrow \sin \cos \sin 1 \geq \tan^{-1}x \geq \sin \cos 1$$

$$\Rightarrow \tan \sin \cos \sin 1 \geq x \geq \tan \sin \cos 1$$

$$21. \text{ Given } 3x - x^2 \geq 2 \Rightarrow x^2 - 3x + 2 \leq 0 \Rightarrow x = 1 \text{ (or) } 2$$

At $x=2$ sine is not defined

So $\sin^{-1}(1) + \sin^{-1}(1) + \dots + 10$ times

$$= \frac{\pi}{2} + \frac{\pi}{2} + \dots + 10 \text{ times} = 5\pi$$

$$22. \quad \tan^{-1}x - \tan^{-1}y = 0 \Rightarrow x = y$$

$$\cos^{-1}x + \cos^{-1}y = \frac{\pi}{2} \Rightarrow x = \frac{1}{\sqrt{2}}$$

$$23. \text{ Apply } \cot^{-1}x = \cos^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) \text{ and}$$

$$\cot^{-1}x = \sin^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right) \text{ formulas}$$

$$24. \quad x^2 - 2x + 2 = (x-1)^2 + 1 \geq 1 \therefore x = 1$$

$$a + \frac{\pi}{2} = 0 \Rightarrow a = -\frac{\pi}{2}$$

$$25. \quad (\sin^{-1}x)^2 + (\cos^{-1}x)^2 = (\sin^{-1}x)^2 + \left(\frac{\pi}{2} - \sin^{-1}x\right)^2$$

$$= 2(\sin^{-1}x)^2 - \pi \sin^{-1}x + \frac{\pi^2}{4}$$

$$\therefore \text{ Its minimum value is } \frac{4.2. \frac{\pi^2}{4} - \pi^2}{8} = \frac{\pi^2}{8}$$

$$26. (\tan^{-1}x + \cot^{-1}x)^2 - 2\tan^{-1}x\left(\frac{\pi}{2} - \tan^{-1}x\right) = \frac{5\pi^2}{8}$$

$$2(\tan^{-1}x)^2 - \pi\tan^{-1}x - \frac{3\pi^2}{8} = 0$$

$$\Rightarrow \tan^{-1}x = -\frac{\pi}{4}, \frac{3\pi}{4}$$

$$\Rightarrow \tan^{-1}x = -\frac{\pi}{4} \Rightarrow x = -1$$

$$27. \sin^{-1}(x^2 - 6x + 12) = \frac{\pi}{3}$$

$$x^2 - 6x + 12 = (x-3)^2 + 3 \geq 3 \quad \forall x \in \mathbb{R}$$

$$28. \text{ If } -1 \leq x < \frac{1}{\sqrt{2}} \Rightarrow \cos^{-1}x > \sin^{-1}x$$

$$29. 2 \tan^{-1}x > \frac{\pi}{2}.$$

$$30. \sin^{-1}(\sin 12) + \cos^{-1}(\cos 12) = 12 - 4\pi + 4\pi - 12 = 0$$

$$31. \text{ Apply } \cos^{-1}x - \cos^{-1}y = \cos^{-1}(xy + \sqrt{1-x^2}\sqrt{1-y^2})$$

$$32. \log_2 x \geq 0 \Rightarrow 1, \text{ for}$$

$$x \geq 1, \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \pi - 2\tan^{-1}x$$

$$33. \sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \frac{3\pi}{2} \Rightarrow x = y = z = 1$$

$$34. \text{ common domain of P, Q, R is } \{-1, 1\}$$

$$\text{min of P is at } x = -1 \Rightarrow P = -\frac{3\pi}{2}$$

$$\text{min of Q is at } x = -1 \Rightarrow Q = \frac{\pi}{2}$$

$$\text{min of R is at } x = 1 \Rightarrow R = -\frac{\pi}{2}$$

$$35. \sum_{n=1}^n \tan^{-1}\left(\frac{1}{1+(n-1)n}\right) = \tan^{-1}(n) - \tan^{-1}(1)$$



1. Range of $\sin^{-1}x - \cos^{-1}x$ is

$$1) \left[-\frac{3\pi}{2}, \frac{\pi}{2}\right] \quad 2) \left[-\frac{5\pi}{3}, \frac{\pi}{3}\right]$$

$$3) \left[-\frac{3\pi}{2}, \pi\right] \quad 4) [0, \pi]$$

2. The ascending order of $A = \sin^{-1}(\log_3^2)$,

$$B = \cos^{-1}\left(\log_3\left(\frac{1}{2}\right)\right), C = \tan^{-1}(\log_{1/3} 2) \text{ is}$$

$$1) C, B, A \quad 2) B, A, C \quad 3) C, A, B \quad 4) B, C, A$$

3. The domain of $\tan^{-1}(5x)$

$$1) \mathbb{R} \quad 2) (0, \pi) \quad 3) [0, \pi] \quad 4) \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$$

4. The decreasing order of, $A = \left(\sin^{-1}\left(\frac{1-\sqrt{3}}{2\sqrt{2}}\right)\right)$

$$B = \cos^{-1}\left(\frac{-1}{\sqrt{2}}\right), C = \tan^{-1}\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right) \text{ is}$$

$$1) B, A, C \quad 2) B, C, A \quad 3) C, A, B \quad 4) C, B, A$$

5. The value of $\sin^{-1}(\cos(\frac{33\pi}{5}))$ is

$$1) \frac{3\pi}{5} \quad 2) \frac{7\pi}{5} \quad 3) \frac{\pi}{10} \quad 4) \frac{-\pi}{10}$$

$$6. \cos^{-1}\left(\cos \frac{10\pi}{7}\right) + \sin^{-1}\left(\sin \frac{35\pi}{11}\right) +$$

$$\tan^{-1}\left(\tan \frac{24\pi}{13}\right) + \cot^{-1}\left(\cot \frac{26\pi}{5}\right) \text{ is}$$

$$1) \frac{4\pi}{7} + \frac{\pi}{55} + \frac{2\pi}{13} \quad 2) \frac{4\pi}{7} - \frac{\pi}{55} - \frac{2\pi}{13}$$

$$3) \frac{4\pi}{7} - \frac{\pi}{55} + \frac{2\pi}{13} \quad 4) -\frac{4\pi}{7} + \frac{\pi}{55} + \frac{2\pi}{13}$$

7. The value of $\tan\left(\frac{1}{2}\cos^{-1}\left(\frac{\sqrt{5}}{3}\right)\right)$ is

$$1) \frac{3+\sqrt{5}}{2} \quad 2) 3+\sqrt{5}$$

$$3) \frac{1}{2}(3-\sqrt{5}) \quad 4) 2-\sqrt{3}$$

8. If $\tan^{-1}\frac{5}{12} + \tan^{-1}\frac{24}{25} = \cos^{-1}x$ then $x =$

$$1) \frac{-31}{325} \quad 2) \frac{-33}{325} \quad 3) \frac{-36}{325} \quad 4) \frac{-39}{325}$$

9. $6\sin^{-1}(x^2 - 6x + 8.5) = \pi$, then $x =$

$$1) 1 \quad 2) 2 \quad 3) 3 \quad 4) 0$$

10. If $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1} \frac{8}{31}$ then $x =$
 1) 1 2) $1/2$ 3) $-1/2$ 4) $1/4$

11. If $\cot^{-1}\left(\frac{1}{x+1}\right) + \cot^{-1}\left(\frac{1}{x-1}\right) = \tan^{-1} 3x - \tan^{-1} x$ then $x =$
 1) $\pm 1/2$ 2) $-1, \pm 1/3$ 3) $2, \pm 1$ 4) $-1, \pm 1/2$

12. $\tan^{-1} \frac{5}{6} + \frac{1}{2} \tan^{-1} \frac{11}{60} =$
 1) π 2) $\pi/2$ 3) $\pi/4$ 4) $3\pi/4$

13. $\tan\left(\frac{\pi}{4} + \frac{1}{2} \cos^{-1} \frac{a}{b}\right) + \tan\left(\frac{\pi}{4} - \frac{1}{2} \cos^{-1} \frac{a}{b}\right)$
 1) b/a 2) a/b 3) $2a/b$ 4) $2b/a$

14. $\cos^{-1}\left\{\frac{1}{\sqrt{2}}\left(\cos \frac{9\pi}{10} - \sin \frac{9\pi}{10}\right)\right\} =$
 1) $\frac{3\pi}{20}$ 2) $\frac{7\pi}{10}$ 3) $\frac{7\pi}{20}$ 4) $\frac{17\pi}{20}$

15. If $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$ then $x =$
 1) $\frac{1}{\sqrt{2}}$ 2) $\pm \frac{1}{\sqrt{2}}$ 3) $\pm \frac{1}{\sqrt{3}}$ 4) $\frac{1}{\sqrt{3}}$

16. The number of solutions of the equation $2(\sin^{-1} x)^2 - 5\sin^{-1} x + 2 = 0$ is
 1) 0 2) 1 3) 2 4) 3

17. If $\frac{1}{\sqrt{2}} < x < 1$ then

$$\cos^{-1} x + \cos^{-1}\left(\frac{x + \sqrt{1-x^2}}{\sqrt{2}}\right) =$$

1) $2\cos^{-1} x - \frac{\pi}{4}$ 2) $2\cos^{-1} x$ 3) $\frac{\pi}{4}$ 4) 0

18. Value of $\tan^{-1} 2 + \tan^{-1} 3 + \tan^{-1} 4 =$

1) $\pi - \tan^{-1} \frac{3}{5}$ 2) $\pi + \tan^{-1} \frac{5}{3}$
 3) $\pi - \tan^{-1} \frac{5}{3}$ 4) $\pi + \tan^{-1} \frac{3}{5}$

19. The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in

1) $(\frac{\pi}{4}, \frac{\pi}{2})$ 2) $(-\frac{\pi}{2}, \frac{\pi}{4})$ 3) $(0, \frac{\pi}{2})$ 4) $(-\frac{\pi}{2}, \frac{\pi}{2})$

20. If $[\cot^{-1} x] + [\cos^{-1} x] = 0$ where x is non-negative real number and $[\bullet]$ denotes the greatest integer function, then complete set of values of x is

1) $(\cos 1, 1]$ 2) $(\cos 1, \cot 1)$
 3) $(\cot 1, 1]$ 4) $(0, \cos 1)$

21. For the equation $\cos^{-1} x + \cos^{-1} 2x + \pi = 0$ the number of real solution is

1) 0 2) 1 3) 2 4) 3

22. If $\alpha = 2 \tan^{-1}(\sqrt{2} - 1),$

$\beta = 3 \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$ and

$\gamma = \cos^{-1} \frac{1}{3}$ then

1) $\alpha < \beta < \gamma$ 2) $\alpha < \gamma < \beta$
 3) $\beta < \gamma < \alpha$ 4) $\gamma < \beta < \alpha$

23. The value of

$\sin^{-1}\left(\cot\left(\sin^{-1}\sqrt{\frac{2-\sqrt{3}}{4}} + \cos^{-1}\frac{\sqrt{12}}{4} + \sec^{-1}\sqrt{2}\right)\right)$ is
 1) 0 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ 4) π

24. The equation $2\cos^{-1} x + \sin^{-1} x = \frac{11\pi}{6}$ has

1) No solution 2) One solution
 3) Two solutions 3) Three solutions

25. The smallest and the largest values of

$\tan^{-1}\left(\frac{1-x}{1+x}\right), 0 \leq x \leq 1$ are

1) $0, \pi$ 2) $0, \frac{\pi}{4}$ 3) $-\frac{\pi}{4}, \frac{\pi}{4}$ 4) $\frac{\pi}{4}, \frac{\pi}{2}$

26. If α, β, γ are the roots of the equation $x^3 + mx^2 + 3x + m = 0$, then the general value of $\tan^{-1} \alpha + \tan^{-1} \beta + \tan^{-1} \gamma$ is

1) $(2n+1)\frac{\pi}{2}$ 2) $n\pi$ 3) $\frac{n\pi}{2}$
 4) dependent upon the value of m

27. For $0 \leq \cos^{-1} x \leq \pi$ and $-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$ then

value of $\cos(\sin^{-1} x + 2\cos^{-1} x)$ at $x = \frac{1}{5}$ is

1) $\frac{-2\sqrt{6}}{5}$ 2) $\frac{-\sqrt{6}}{5}$ 3) $\frac{2\sqrt{6}}{5}$ 4) $\frac{\sqrt{6}}{5}$

28. If 'a' is twice the tangent of the arithmetic mean of $\sin^{-1}x$ and $\cos^{-1}x$, 'b' is the geometric mean of $\tan x$ and $\cot x$. Then

$$x^2 - ax + b = 0 \Rightarrow x =$$

- 1) 2 2) 3 3) 1 4) 0

29. If $x > 1$ then $2 \tan^{-1}x$ cannot be equal to

1) $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ 2) $\pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right)$

3) $\pi + \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ 4) $\pi - \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

30. $\cos^{-1}(\cos 12) - \sin^{-1}(\sin 12) =$

- 1) 0 2) π 3) $8\pi + 24$ 4) $8\pi - 24$

31. If $\cos^{-1}\frac{x}{a} + \cos^{-1}\frac{y}{b} = \frac{5\pi}{12}$ and

$$\sin^{-1}\frac{x}{a} - \sin^{-1}\frac{y}{b} = \frac{\pi}{12} \text{ then } \frac{x^2}{a^2} + \frac{y^2}{b^2} =$$

- 1) 1 2) $\frac{1}{4}$ 3) $\frac{3}{4}$ 4) $\frac{5}{4}$

32. $2\cos^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$ is valid for

- 1) $-1 \leq x \leq 1$ 2) $0 \leq x \leq 1$

- 3) $0 \leq x \leq \frac{1}{\sqrt{2}}$ 4) $\frac{1}{\sqrt{2}} \leq x \leq 1$

33. If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \frac{\pi}{2}$ and

$$(x-y)^2 + (y-z)^2 + (z-x)^2 = 0 \text{ then}$$

$$x^2 + y^2 + z^2 =$$

- 1) 0 2) 4 3) 1 4) 2

34. The ascending order of $A = \cos(\sin^{-1}\frac{12}{13})$

$$B = \sin(2\tan^{-1}\frac{3}{4}) \quad C = \tan(\frac{1}{2}\cos^{-1}\frac{1}{3}) \text{ is}$$

- 1) A, B, C 2) A, C, B 3) B, C, A 4) B, A, C

35. $\sum_{m=1}^n \tan^{-1}\left(\frac{2m}{m^4 + m^2 + 2}\right) =$

1. $\tan^{-1}\left(\frac{n^2 + n}{n^2 + n + 2}\right)$ 2. $\tan^{-1}\left(\frac{n^2 - n}{n^2 - n + 2}\right)$

3. $\tan^{-1}\left(\frac{n^2 + n + 2}{n^2 + n}\right)$ 4. $\frac{\pi}{4}$

LEVEL - II (H.W.)-KEY

- 01) 1 02) 3 03) 1 04) 2 05) 4 06) 1
07) 3 08) 3 09) 2 10) 4 11) 1 12) 3
13) 4 14) 4 15) 2 16) 2 17) 3 18) 4
19) 2 20) 3 21) 1 22) 2 23) 1 24) 1
25) 2 26) 2 27) 1 28) 3 29) 4 30) 4
31) 4 32) 4 33) 3 34) 2 35) 1

LEVEL - II (H.W.)-HINTS

1. Let $y = \sin^{-1}x - \cos^{-1}x = \frac{\pi}{2} - 2\cos^{-1}x$

$$0 \leq \cos^{-1}x \leq \pi \Rightarrow -\frac{3\pi}{2} \leq y \leq \frac{\pi}{2}$$

2. $0 < A < 90^\circ, B > 90^\circ, C < 0$

3. Domain of $\tan^{-1}x$ is $\mathbb{R}$

4. $A < 0, B > 90^\circ, 0 < C < 90^\circ$

5. $\frac{33\pi}{5} = 6\pi + \frac{3\pi}{5} \in Q1$

$$\sin^{-1}[\cos(\frac{33\pi}{5})] = \sin^{-1}[\cos(\frac{3\pi}{5})]$$

6. $\cos^{-1}\cos\left(\frac{10\pi}{7}\right) =$

$$\cos^{-1}\cos\left(2\pi - \frac{10\pi}{7}\right) = \cos^{-1}\cos\left(\frac{4\pi}{7}\right) = \left(\frac{4\pi}{7}\right)$$

$$\sin^{-1}\sin\left(\frac{35\pi}{11}\right) = \sin^{-1}\sin\left(2\pi + \frac{13\pi}{11}\right) +$$

$$\sin^{-1}\sin\left(\frac{13\pi}{11}\right) = \sin^{-1}\sin\left(\pi + \frac{2\pi}{11}\right)$$

$$= \sin^{-1}\sin\left(-\frac{2\pi}{11}\right) = -\frac{2\pi}{11}$$

$$\tan^{-1}\tan\left(\frac{24\pi}{13}\right) = \tan^{-1}\tan\left(\frac{2\pi}{13} - 2\pi\right) = \tan^{-1}$$

$$\tan\left(\frac{2\pi}{13}\right) = \frac{2\pi}{13}]$$

$$\cot^{-1}\left(\cot\frac{26\pi}{5}\right) = \cot^{-1}\left(\cot\left(5\pi + \frac{\pi}{5}\right)\right) = \frac{\pi}{5}$$

7. Put $\cos^{-1}\frac{\sqrt{5}}{3} = x$

$$8. \quad \sin^{-1}x = \tan^{-1} \frac{x}{\sqrt{1-x^2}} \Rightarrow \sin^{-1} \frac{24}{25} = \tan^{-1} \frac{24}{7}$$

$$\text{Apply } \tan^{-1}x + \tan^{-1}y = x + \tan^{-1} \left(\frac{x+y}{1-xy} \right),$$

$xy > 1$ formula

$$9. \quad 6 \sin^{-1} \left(x^2 - 6x + \frac{17}{2} \right) = \pi$$

$$\sin^{-1} \left(x^2 - 6x + \frac{17}{2} \right) = \frac{\pi}{6}$$

$x^2 - 6x + \frac{17}{2} = \frac{1}{2}$ solve the equation. It has two values.

10. Apply $\tan^{-1}x + \tan^{-1}y$ formula

$$11. \quad \tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}3x - \tan x$$

Apply $\tan^{-1}x + \tan^{-1}y$ and

$\tan^{-1}x - \tan^{-1}y$ formula

$$12. \quad \frac{1}{2} [2 \tan^{-1} \frac{5}{6} + \tan^{-1} \frac{11}{60}] = \frac{1}{2} [\tan^{-1} \frac{60}{11} + \tan^{-1} \frac{11}{60}] = \frac{1}{2} \pi = \frac{\pi}{2}$$

$$13. \quad \text{Put } \frac{1}{2} \cos^{-1} \frac{a}{b} = \theta \Rightarrow \cos 2\theta = \frac{a}{b}$$

$$\text{Apply } \tan \left(\frac{\pi}{4} + \theta \right) + \tan \left(\frac{\pi}{4} - \theta \right) = \frac{2}{\cos 2\theta}$$

$$14. \quad \cos^{-1} \left[\cos \left(\frac{9\pi}{10} + \frac{\pi}{4} \right) \right] = \cos^{-1} \left[\cos \left(\frac{23\pi}{20} \right) \right]$$

$$= \cos^{-1} \left[\cos \left(2\pi - \frac{17\pi}{20} \right) \right]$$

$$= \cos^{-1} \left[\cos \left(\frac{17\pi}{20} \right) \right] = \frac{17\pi}{20}$$

15. Apply $\tan^{-1}x + \tan^{-1}y$ formula

16. Put $\sin^{-1}x = a$

$$17. \quad \cos^{-1}x + \cos^{-1} \left(x \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \sqrt{1-x^2} \right)$$

$$= \cos^{-1}x + \cos^{-1} \left(\frac{1}{\sqrt{2}} \right) - \cos^{-1}x = \frac{\pi}{4}$$

$$18. \quad \text{Apply } \tan^{-1}x + \tan^{-1}y = \pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right), xy > 1$$

And $\tan^{-1}x - \tan^{-1}y$ formula

$$19. \quad f(x) = \tan^{-1}(\sin x + \cos x) \text{ find } f'(x)$$

$$f'(x) > 0 \text{ if } -\frac{\pi}{2} < x + \frac{\pi}{4} < \frac{\pi}{2} \text{ then}$$

$$-\frac{3\pi}{4} < x < \frac{\pi}{4} \text{ Hence } f(x) \text{ is increasing}$$

$$\text{when } x \in \left(-\frac{\pi}{2}, \frac{\pi}{4} \right)$$

$$20. \quad 0 < \cot^{-1}x < \pi \text{ and } 0 \leq \cos^{-1}x \leq \pi$$

$$\cot^{-1} 1 < x < \infty \text{ and } \cos^{-1} 1 \leq x \leq 1$$

$$x \in (\cot^{-1} 1, 1] \because \cos^{-1} 1 < \cot^{-1} 1$$

21. We have $\cos^{-1}x + \cos^{-1}2x = -\pi$ which is not possible as $\cos^{-1}x$ and $\cos^{-1}2x$ never take negative values

$$22. \quad \alpha = 2 \tan^{-1}(\sqrt{2}-1) = \frac{\pi}{4}, \beta = \frac{7\pi}{12},$$

$$\gamma = \cos^{-1} \left(\frac{1}{3} \right) > \frac{\pi}{4}$$

$$23. \quad \sin^{-1} \left[\cot \left(\frac{\pi}{12} + \frac{\pi}{6} + \frac{\pi}{4} \right) \right] = \sin^{-1} \left[\cot \left(\frac{\pi}{2} \right) \right]$$

$$= \sin^{-1}(0) = 0$$

$$\therefore \sin^{-1}x \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

$$24. \quad \cos^{-1}x + (\cos^{-1}x + \sin^{-1}x) = \frac{11\pi}{6} \Rightarrow \cos^{-1}x + \frac{\pi}{2} = \frac{11\pi}{6}$$

$$\Rightarrow \cos^{-1}x = \frac{4\pi}{3} \text{ which is not possible as}$$

$$\cos^{-1}x \in [0, \pi]$$

$$25. \quad \text{We have } \tan^{-1} \left(\frac{1-x}{1+x} \right) = \frac{\pi}{4} - \tan^{-1}x$$

$$\text{Since } 0 \leq x \leq 1 \Rightarrow 0 \leq \tan^{-1}x \leq \frac{\pi}{4} \Rightarrow 0 \leq \tan^{-1} \left(\frac{1-x}{1+x} \right) \leq \frac{\pi}{4}$$

$$26. \quad \text{Let } \alpha = \tan \theta_1, \beta = \tan \theta_2, \gamma = \tan \theta_3$$

$$\tan(\theta_1 + \theta_2 + \theta_3) \Rightarrow \theta_1 + \theta_2 + \theta_3 = n\pi, n \in \mathbb{Z}$$

$$27. \quad \cos \left(\frac{\pi}{2} + \cos^{-1}x \right) = -\sqrt{1-x^2}$$

$$28. \quad a = 2 \tan \left(\frac{\sin^{-1}x + \cos^{-1}x}{2} \right) = 2(1) = 2$$

$$b = \sqrt{\tan x \cdot \cot x} = 1$$

$$\therefore x^2 - ax + b = 0 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow x = 1$$

29. if $x > 1$ then $2 \tan^{-1} x > \frac{\pi}{2}$ by verification

$$\pi - \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) \text{ is not equal to } 2 \tan^{-1} x.$$

30. $(4\pi - 12) - (12 - 4\pi) = 8\pi - 24$

31. $\cos^{-1} \frac{x}{a} + \cos^{-1} \frac{y}{b} = \frac{5\pi}{2} \rightarrow (1) \text{ and}$

$$\cos^{-1} \frac{y}{b} - \cos^{-1} \frac{x}{a} = \frac{\pi}{2} \rightarrow (2)$$

$$(1)+(2) \Rightarrow \cos^{-1} \frac{y}{b} = \frac{\pi}{4} \Rightarrow \frac{y}{b} = \frac{1}{\sqrt{2}}$$

$$(1)-(2) \Rightarrow \cos^{-1} \frac{x}{a} = \frac{\pi}{6} \Rightarrow \frac{x}{a} = \frac{\sqrt{3}}{2}$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{5}{4}$$

32. Let $x = \cos y$ where $0 \leq y \leq \pi$, $|x| \leq 1$

$$2 \cos^{-1} x = \sin^{-1} (2x\sqrt{1-x^2}) \rightarrow (1)$$

$$\Rightarrow 2 \cos^{-1} (\cos y) = \sin^{-1} (2 \cos y \sqrt{1 - \cos^2 y})$$

$$\Rightarrow 2 \cos^{-1} (\cos y) = \sin^{-1} (\sin 2y)$$

$$\sin^{-1} (\sin 2y) = 2y \text{ for } -\frac{\pi}{4} \leq y \leq \frac{\pi}{4}$$

$$\text{And } 2 \cos^{-1} (\cos y) = 2y \text{ for } 0 \leq y \leq \pi$$

Thus equation (1) holds only when

$$y \in [0, \frac{\pi}{4}] \Rightarrow x \in [\frac{1}{\sqrt{2}}, 1]$$

33. Put $x = y = z = \frac{1}{\sqrt{3}}$,

$$x^2 + y^2 + z^2 = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$$

34. $A = \cos(\sin^{-1} \frac{12}{13}) = \frac{5}{13}$

$$(\sin^{-1} x = \cos^{-1} \sqrt{1-x^2})$$

$$B = \sin(2 \tan^{-1} \frac{3}{4}) = \frac{24}{25} \left(\because 2 \tan^{-1} x = \sin^{-1} \left(\frac{2x}{1+x^2} \right) \right)$$

$$C = \tan\left(\frac{1}{2} \cos^{-1} \frac{1}{3}\right) \text{ put } \cos^{-1} \frac{1}{3} = \theta$$

$$\text{then } C = \frac{1}{\sqrt{2}}$$

35. $\sum_{m=1}^n \tan^{-1} \left(\frac{2m}{m^4 + m^2 + 2} \right)$

$$= \sum_{m=1}^n \tan^{-1} \left(\frac{(m^2 + m + 1) - (m^2 - m + 1)}{1 + (m^2 + m + 1)(m^2 - m + 1)} \right)$$

$$= \sum_{m=1}^n [\tan^{-1} (m^2 + m + 1) - \tan^{-1} ((m^2 - m + 1))]$$



LEVEL - III

1. $\frac{\cos^{-1} (41/49)}{\sin^{-1} (2/7)} =$

1) 4 2) 3 3) 2 4) 1

2. If x takes negative permissible value, then

$$\sin^{-1} x =$$

1) $\cos^{-1} \sqrt{1-x^2}$ 2) $-\cos^{-1} \sqrt{1-x^2}$
3) $\cos^{-1} \sqrt{x^2-1}$ 4) $\pi - \cos^{-1} \sqrt{1-x^2}$

3. $\cos^{-1} \sqrt{\frac{a-x}{a-b}} = \sin^{-1} \sqrt{\frac{x-b}{a-b}}$ ($a \neq b$) is possible if

1) $a \geq x \geq b$ or $a \leq x \leq b$
2) $a = x = b$
3) $a > b$ and x takes any value
4) $a < b$ and x takes any value

4. The value of $\sin^{-1} \left(\cos \left\{ \cos^{-1} (\cos x) + \sin^{-1} (\sin x) \right\} \right)$

where $x \in \left(\frac{\pi}{2}, \pi \right)$ is

1) $\frac{\pi}{2}$ 2) $-\pi$ 3) π 4) $-\frac{\pi}{2}$

5. If $x = \sin(2 \tan^{-1} 2)$ and $y = \sin \left[\frac{1}{2} \tan^{-1} \frac{4}{3} \right]$ then

1) $x = y$ 2) $x > y$ 3) $\frac{2}{\sqrt{5}}$ 4) $x < y$

6. $\sin^{-1}(\sin 3) + \cos^{-1}(\cos 7) - \tan^{-1}(\tan 5) =$

1) $\pi - 1$ 2) π 3) $3\pi - 1$ 4) $2\pi - 1$

7. The number of solutions of

$$\sin^{-1} (1 + b + b^2 + \dots \infty) + \cos^{-1} \left(a - \frac{a^2}{3} + \frac{a^3}{9} + \dots \infty \right) = \frac{\pi}{2}$$

1) 1 2) 2 3) 3 4) ∞

INVERSE TRIGONOMETRIC FUNCTIONS

8. If $y = (\sin^{-1}x)^3 + (\cos^{-1}x)^3$ then

- 1) $\min y = \frac{\pi^3}{8}$ 2) $\min y = \frac{\pi^3}{32}$
 3) $\max y = \frac{\pi^3}{8}$ 4) $\max y = \frac{7\pi^3}{32}$

9. $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + \sin^{-1}\left(\frac{\sqrt{2}-1}{\sqrt{6}}\right) + \dots + \sin^{-1}\left(\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}}\right) + \dots =$

- 1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{3\pi}{2}$

10. $\cos^{-1}\left\{\frac{1}{2}x^2 + \sqrt{1-x^2}\sqrt{1-\frac{x^2}{4}}\right\} = \cos^{-1}\frac{x}{2} - \cos^{-1}x$

holds for :

- 1) $|x| \leq 1$ 2) $x \in R$
 3) $0 \leq x \leq 1$ 4) $-1 \leq x \leq 0$

11. The set of values of x from which the identity

$$\cos^{-1}x + \cos^{-1}\left(\frac{x}{2} + \frac{1}{2}\sqrt{3-3x^2}\right) = \frac{\pi}{3} \quad \text{holds good}$$

is

- 1) $[0, 1]$ 2) $[0, \frac{1}{2}]$ 3) $[\frac{1}{2}, 1]$ 4) $\{-1, 0, 1\}$

12. If $A = \tan^{-1}\left(\frac{x\sqrt{3}}{2k-x}\right)$ and $B = \tan^{-1}\left(\frac{2x-k}{k\sqrt{3}}\right)$ then $A-B =$

- 1) 0° 2) $\pi/6$ 3) $\pi/4$ 4) $\pi/3$

13. The number of solutions of the equation

$$\tan^{-1}(x-1) + \tan^{-1}(x) + \tan^{-1}(x+1) = \tan^{-1}(3x) \text{ is}$$

- 1) 1 2) 2 3) 3 4) 4

14. $\cot^{-1}(\sqrt{\cos\alpha}) - \tan^{-1}(\sqrt{\cos\alpha}) = x \geq 0$, then $\sin x =$ (AIE-2002)

- 1) $\tan^2 \frac{\alpha}{2}$ 2) $\cot^2 \left(\frac{\alpha}{2}\right)$ 3) $\tan \alpha$ 4) $\cot \frac{\alpha}{2}$

15. If $a_1, a_2, a_3, \dots, a_n$ are in A.P. with common difference 'd', then

$$\tan \left[\tan^{-1}\left(\frac{d}{1+a_1a_2}\right) + \tan^{-1}\left(\frac{d}{1+a_2a_3}\right) + \dots + \tan^{-1}\left(\frac{d}{1+a_{n-1}a_n}\right) \right] =$$

- 1) $\frac{(n-1)d}{a_1+a_n}$ 2) $\frac{(n-1)d}{1+a_1a_n}$

MATHS BY Er. RG SIR

- 3) $\frac{nd}{1+a_1a_n}$ 4) $\frac{a_n - a_1}{a_n + a_1}$

16. $\tan^{-1}\left(\frac{c_1x-y}{c_1y+x}\right) + \tan^{-1}\left(\frac{c_2-c_1}{1+c_2c_1}\right) + \tan^{-1}\left(\frac{c_3-c_2}{1+c_3c_2}\right) + \dots + \tan^{-1}\left(\frac{1}{c_n}\right) =$

- 1) $\tan^{-1}\left(\frac{2x}{y}\right)$ 2) $\tan^{-1}(xy)$

- 3) $\tan^{-1}\left(\frac{x}{y}\right)$ 4) $\tan^{-1}\left(\frac{y}{x}\right)$

17. $\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{2}{9}\right) + \tan^{-1}\left(\frac{4}{33}\right) + \dots =$

- 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) π 4) 2π

18. Value of $\sec^{-1}\left(\frac{1}{1-2x^2}\right) + 4\cos^{-1}\sqrt{\frac{1+x}{2}} =$

- 1) $2\tan^{-1}x$ 2) $\tan^{-1}\left(\frac{\sqrt{1-x^2}}{1-x}\right)$

- 3) $\cot^{-1}\left(\frac{\sqrt{1-x^2}}{1-x}\right)$ 4) constant for all x

19. The solution set of the equation

$$\tan^{-1}x - \cot^{-1}x = \cos^{-1}(2-x) \text{ is :}$$

- 1) $[0, 1]$ 2) $[-1, 1]$ 3) $[1, 3]$ 4) $(1, 3)$

20. The number of solutions of the equation

$$3\cos^{-1}x - \pi x - \frac{\pi}{2} = 0$$

- 1) 0 2) 1 3) 2 4) infinitely many

21. Number of real solutions of the equation

$$\sqrt{1+\cos 2x} = \sqrt{2} \sin^{-1}(\sin x) \text{ where } -\pi \leq x \leq \pi$$

- 1) 0 2) 1 3) 2 4) 4

22. If the equation

$$\sin^{-1}(x^2 + x + 1) + \cos^{-1}(\lambda x + 1) = \frac{\pi}{2}$$

has exactly two solutions, then λ can have the integral value.

- 1) -1 2) 0 3) 1 4) 2

23. If x_1, x_2, x_3, x_4 are roots of the equation

$$x^4 - x^3 \sin 2\beta + x^2 \cos 2\beta - x \cos \beta - \sin \beta = 0$$

then $\tan^{-1} x_1 + \tan^{-1} x_2 + \tan^{-1} x_3 + \tan^{-1} x_4 =$

- 1) β 2) $\frac{\pi}{2} - \beta$ 3) $\pi - \beta$ 4) $-\beta$

24. If $\cot^{-1} \frac{n}{\pi} > \frac{\pi}{6}, n \in \mathbb{N}$ then the maximum value of 'n' is

- 1) 6 2) 7 3) 5 4) 10

25. If $(\cot^{-1} x)^2 - 7(\cot^{-1} x) + 10 > 0$, then x lies in the interval

- 1) $(\cot 5, \cot 2)$ 2) $(-\infty, \cot 5) \cup (\cot 2, \infty)$
3) $(-\infty, \cot 5)$ 4) $(\cot 2, \infty)$

26. The value of x for which

$$\cos^{-1}(\cos 4) > 3x^2 - 4x \text{ is}$$

- 1) $\left(0, \frac{2+\sqrt{6\pi-8}}{3}\right)$ 2) $\left(\frac{2+\sqrt{6\pi-8}}{3}, 0\right)$
3) $(-2, 2)$ 4) $\left(\frac{2-\sqrt{6\pi-8}}{3}, \frac{2+\sqrt{6\pi-8}}{3}\right)$

27. The least integral value of k for which $(k-2)x^2 + 8x + k + 4 > \sin^{-1}(\sin 12) + \cos^{-1}(\cos 12)$ for all $x \in \mathbb{R}$, is

- 1) -7 2) -5 3) -3 4) 5

28. Point $P(x, y)$ satisfying the equation $\sin^{-1} x +$

$$\cos^{-1} y + \cos^{-1}(2xy) = \frac{\pi}{2} \text{ lies on}$$

- 1) the bisector of the first and third quadrant
2) bisector of the second and fourth quadrant.
3) the rectangle formed by the lines $x = \pm 1$ and $y = \pm 1$.
4) a unit circle with centre at the origin.

29. If α is the only real root of the equation

$$x^3 + bx^2 + cx + 1 = 0 (b < c) \text{ then the value of}$$

$$\tan^{-1} \alpha + \tan^{-1} \left(\frac{1}{\alpha}\right) =$$

- 1) $\frac{\pi}{2}$ 2) $-\frac{\pi}{2}$ 3) 0 4) π

30. $\frac{\alpha^3}{2} \operatorname{cosec}^2 \left(\frac{1}{2} \tan^{-1} \frac{\alpha}{\beta}\right) +$

$$\frac{\beta^3}{2} \sec^2 \left(\frac{1}{2} \tan^{-1} \left(\frac{\beta}{\alpha}\right)\right) =$$

- 1) $(\alpha - \beta)(\alpha^2 + \beta^2)$ 2) $(\alpha + \beta)(\alpha^2 - \beta^2)$
3) $(\alpha + \beta)(\alpha^2 + \beta^2)$ 4) 0

31. $x = n\pi - \tan^{-1} 3$ is a solution of the equation

$$12 \tan 2x + \frac{\sqrt{10}}{\cos x} + 1 = 0 \text{ if}$$

- 1) n is any integer
2) n is an even integer
3) n is a positive integer
4) n is an odd integer

LEVEL - III-KEY

- 01) 3 02) 2 03) 1 04) 4 05) 2 06) 1
07) 4 08) 2 09) 2 10) 3 11) 3 12) 2
13) 3 14) 1 15) 2 16) 3 17) 1 18) 4
19) 3 20) 2 21) 3 22) 2 23) 2 24) 3
25) 4 26) 4 27) 4 28) 4 29) 2 30) 3
31) 4

LEVEL - III-HINTS

1. $\cos^{-1} \frac{41}{49} = 2 \sin^{-1} \left(\frac{2}{7}\right)$

2. Let $\sin^{-1} x = y \Rightarrow x = \sin y$

since $-1 \leq x \leq 0$ therefore $-\frac{\pi}{2} \leq \sin^{-1} x \leq 0$

$$\Rightarrow \frac{\pi}{2} \geq -y \geq 0$$

$$\Rightarrow (-y) = \cos^{-1} \sqrt{1-x^2} \Rightarrow y = -\cos^{-1} \sqrt{1-x^2}$$

3. Take $\sqrt{\frac{a-x}{a-b}} = \sqrt{\frac{x-b}{a-b}} = \frac{1}{\sqrt{2}}$

4. $\sin^{-1} [\cos \{x + \pi - x\}] = -\pi/2$

5. Put $\tan^{-1} 2 = A$ and $\tan^{-1} \frac{4}{3} = B$ then verify

6. $\pi - 3 + 7 - 2\pi - (5 - 2\pi) = \pi - 1$

7. The given equation is valid if $-1 \leq 1 + b + b^2 + \dots \infty \leq 1$, and $a - \frac{a^2}{3} + \frac{a^3}{9} - \dots \infty \in [-1, 1]$

also $1 + b + b^2 + \dots \infty = a - \frac{a^2}{3} + \frac{a^3}{9} - \dots \infty$

8. By standed formula

$$\frac{\pi^3}{32} \leq (\sin^{-1} x)^3 + (\cos^{-1} x)^3 \leq \frac{7\pi^3}{8}$$

$$9. \quad T_n = \sin^{-1} \frac{\sqrt{n} - \sqrt{n-1}}{\sqrt{n(n+1)}} = \tan^{-1} \frac{\sqrt{n} - \sqrt{n-1}}{1 + \sqrt{n}\sqrt{n-1}}$$

$$= \tan^{-1} \sqrt{n} - \tan^{-1} \sqrt{n-1}$$

$$S_\infty = \{\tan^{-1} 1 - \tan^{-1} 0\} + \{\tan^{-1} \sqrt{2} - \tan^{-1} 1\} + \dots + \tan^{-1} \infty$$

$$= \tan^{-1} \infty - \tan^{-1} 0 = \frac{\pi}{2}$$

$$10. \quad \cos^{-1} \left(\frac{1}{2} x^2 + \sqrt{1-x^2} \sqrt{1 - \frac{x^2}{4}} \right)$$

$$= \cos^{-1} \left(\frac{x}{2} x + \sqrt{1-x^2} \sqrt{1 - \left(\frac{x}{2}\right)^2} \right)$$

$$\text{For } \cos^{-1} \left(\frac{1}{2} x^2 + \sqrt{1-x^2} \sqrt{1 - \frac{x^2}{4}} \right)$$

$$= \cos^{-1} \frac{x}{2} - \cos^{-1} x$$

$$\text{L.H.S} > 0, \quad \text{hence} \quad \text{R.H.S} > 0$$

$$\Rightarrow \cos^{-1} \frac{x}{2} - \cos^{-1} x > 0$$

Since $\cos^{-1} x$ is decreasing function

$$\frac{x}{2} \leq x \Rightarrow \frac{x}{2} \geq 0 \Rightarrow x \in [0, 1]$$

$$11. \quad \text{by verification } x = \frac{1}{2} \text{ and } 1$$

$$12. \quad \text{Put } x = k = 1$$

Apply $\tan^{-1} x - \tan^{-1} y$ formula

$$13. \quad \tan^{-1}(x-1) + \tan^{-1}(x+1) = \tan^{-1}(3x) - \tan^{-1} x$$

apply the formula

$$14. \quad \cot^{-1} \sqrt{\cos \alpha} - \tan^{-1} \sqrt{\cos \alpha} = x$$

$$\Rightarrow \frac{\pi}{2} - 2 \tan^{-1} \sqrt{\cos \alpha} = x$$

$$\Rightarrow \frac{\pi}{2} - \tan^{-1} \frac{2\sqrt{\cos \alpha}}{1 - \cos \alpha} = x$$

$$\Rightarrow \cot^{-1} \left(\frac{2\sqrt{\cos \alpha}}{1 - \cos \alpha} \right) = x \Rightarrow \cot x = \frac{2\sqrt{\cos \alpha}}{1 - \cos \alpha}$$

$$\therefore \sin x = \frac{1 - \cos \alpha}{1 + \cos \alpha} = \tan^2 \alpha / 2$$

$$15. \quad \text{Taking terms in numerator as}$$

$$d = a_2 - a_1 = a_3 - a_2 \dots \dots \dots \text{after that apply}$$

$$\tan^{-1} x - \tan^{-1} y \text{ formula}$$

$$16. \quad \tan^{-1} \left(\frac{\frac{x}{y} - \frac{1}{c_1}}{1 + \frac{x}{y} \cdot \frac{1}{c_1}} \right) + \tan^{-1} \left(\frac{\frac{1}{c_1} - \frac{1}{c_2}}{1 + \frac{1}{c_1} \cdot \frac{1}{c_2}} \right) + \dots + \tan^{-1} \left(\frac{1}{c_n} \right)$$

$$\left\{ \tan^{-1} \left(\frac{x}{y} \right) - \tan^{-1} \left(\frac{1}{c_1} \right) \right\} + \left\{ \tan^{-1} \left(\frac{1}{c_1} \right) - \tan^{-1} \left(\frac{1}{c_2} \right) \right\} + \dots + \tan^{-1} \left(\frac{1}{c_n} \right)$$

$$= \tan^{-1} \left(\frac{x}{y} \right)$$

$$17. \quad T_n = \tan^{-1} \left(\frac{2^{n-1}}{1 + 2^{2^{n-1}}} \right) = \tan^{-1} \left(\frac{2^n - 2^{n-1}}{1 + 2^n \cdot 2^{n-1}} \right)$$

$$= \tan^{-1} 2^n - \tan^{-1} 2^{n-1}$$

$$S_\infty = \tan^{-1} \infty - \tan^{-1} 1 = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

$$18. \quad \sec^{-1} \left(\frac{1}{1-2x^2} \right) + 4 \cos^{-1} \sqrt{\frac{1+x}{2}}$$

$$= \cos^{-1} (1-2x^2) + 4 \cos^{-1} \sqrt{\frac{1+x}{2}}$$

$$\therefore \pi - \cos^{-1} (2x^2 - 1) + 4 \cos^{-1} \sqrt{\frac{1+x}{2}} \quad \dots (1)$$

put $x = \cos \theta$ then (1) can be written as

$$\pi - \cos^{-1} (\cos 2\theta) + 4 \cos^{-1} \cos \left(\frac{\theta}{2} \right)$$

$$= \pi - 2\theta + 2\theta = \pi \text{ which is constant for all } x.$$

$$19. \quad \text{Since, } \tan^{-1} x \text{ and } \cot^{-1} x \text{ exists for all } x \in \mathbb{R} \text{ and}$$

$$\cos^{-1} (2-x) \text{ exists, if } -1 \leq 2-x \leq 1$$

$\therefore \tan^{-1} x - \cot^{-1} x = \cos^{-1} (2-x)$ is possible only, if $1 \leq x \leq 3$, thus the solution of given equation is $[1, 3]$ but at $x = 3$ given equation not satisfied therefore $x \in [1, 3)$.

$$20. \quad \cos^{-1} x = \frac{\pi}{3} x + \frac{\pi}{6} \text{ has only one real root.}$$

$$21. \quad \sqrt{1 + \cos 2x} = \sqrt{2} \sin^{-1} (\sin x)$$

$$|\cos x| = \sin^{-1} (\sin x) \text{ The graph of}$$

$$y = |\cos x| \text{ intersect the curve}$$

$$y = \sin^{-1} (\sin x) \text{ in 2 points in } [-\pi, \pi]$$

$$22. \quad x^2 + x + 1 = \lambda x + 1$$

$$\Rightarrow x = 0, \lambda - 1 \text{ and } -1 \leq x \leq 0$$

$$\Rightarrow 0 \leq \lambda < 1 \Rightarrow \lambda = 0$$

23. from given equation.

$$\sum x_1 = \sin 2\beta; \sum x_1 x_2 = \cos 2\beta; \sum x_1 x_2 x_3 = \cos \beta;$$

$$\text{and } x_1 x_2 x_3 x_4 = -\sin \beta$$

$$\therefore \tan^{-1}(x_1) + \tan^{-1} x_2 + \tan^{-1} x_3 + \tan^{-1} x_4 =$$

$$= \tan^{-1} \left[\frac{\sum x_1 - \sum x_1 x_2 x_3}{1 - \sum x_1 x_2 + x_1 x_2 x_3 x_4} \right] = \frac{\pi}{2} - \beta$$

24. $\cot^{-1} \left(\frac{n}{\pi} \right) > \frac{\pi}{6} \Rightarrow \frac{n}{\pi} < \sqrt{3} \quad n < \pi\sqrt{3}$

$$n < 5.46$$

$\therefore$ Maximum value of 'n' is 5.

25. $(\cot^{-1} x)^2 - 7 \cot^{-1} x + 10 > 0$

$$\Rightarrow (\cot^{-1} x - 2)(\cot^{-1} x - 5) > 0$$

Since $\cot^{-1} x - 5$ is negative and does not exits

$$\therefore \cot^{-1} x < 2 \Rightarrow x \in (\cot 2, \infty)$$

26. $\cos^{-1}(\cos 4) = \cos^{-1}\{\cos(2\pi - 4)\} = 2\pi - 4$

$$\Rightarrow 2\pi - 4 > 3x^2 - 4x$$

$$\Rightarrow 3x^2 - 4x - (2\pi - 4) < 0$$

$$\Rightarrow \frac{2 - \sqrt{6\pi - 8}}{3} < x < \frac{2 + \sqrt{6\pi - 8}}{3}$$

27. $\sin^{-1}(\sin 12) = \sin^{-1} \sin(12 - 4\pi) = 12 - 4\pi$

$$\cos^{-1}(\cos 12) = \cos^{-1} \cos(4\pi - 12) = 4\pi - 12$$

$$\setminus (k-2)x^2 + 8x + k + 4 > 0$$

$$\text{if } k = 2 \text{ then } 8x + 4 > 0 \text{ (not possible)}$$

$$\text{if } k > 2 \text{ then } k - 2 > 0$$

$$\text{and } 64 - 4(k-2)(k+4) < 0$$

$$16 < k^2 + 2k - 8, k^2 + 2k - 24 > 0$$

$$(k+6)(k-4) > 0 \Rightarrow k = 5$$

28. $\cos^{-1} y + \cos^{-1}(2xy) = \frac{\pi}{2} - \sin^{-1} x = \cos^{-1} x$

$$\Rightarrow \cos(\cos^{-1}(2xy)) = \cos(\cos^{-1} x - \cos^{-1} y)$$

$$\Rightarrow 2xy = xy + \sqrt{1-x^2} \sqrt{1-y^2} \Rightarrow x^2 + y^2 = 1$$

29. Let $f(x) = x^3 + bx^2 + cx + 1$

$$f(0) = 1 > 0, f(-1) = b - c < 0$$

$$\Rightarrow \alpha \text{ lies between } -1 \text{ and } 0$$

$$\Rightarrow \alpha < 0 \Rightarrow \tan\left(\frac{1}{\alpha}\right) = -\pi + \cot^{-1} \alpha$$

$$\tan^{-1} \alpha + \tan^{-1} \frac{1}{\alpha} = -\pi + \tan^{-1} \alpha + \cot^{-1} \alpha$$

$$= -\pi + \frac{\pi}{2} = -\frac{\pi}{2}$$

30. $\frac{\alpha^3}{2} \cos \sec^2 \left(\frac{1}{2} \tan^{-1} \frac{\alpha}{\beta} \right) + \frac{\beta^3}{2} \sec^2 \left(\frac{1}{2} \tan^{-1} \frac{\beta}{\alpha} \right)$

$$= \alpha^3 \left(\frac{1}{1 - \cos \left(\tan^{-1} \frac{\alpha}{\beta} \right)} \right) + \beta^3 \left(\frac{1}{1 + \cos \left(\tan^{-1} \frac{\beta}{\alpha} \right)} \right)$$

$$= \alpha^3 \left(\frac{1}{1 - \frac{\beta}{\sqrt{\alpha^2 + \beta^2}}} \right) + \beta^3 \left(\frac{1}{1 + \frac{\alpha}{\sqrt{\alpha^2 + \beta^2}}} \right)$$

$$= \sqrt{\alpha^2 + \beta^2} \left\{ \frac{\alpha^3}{\sqrt{\alpha^2 + \beta^2} - \beta} + \frac{\beta^3}{\sqrt{\alpha^2 + \beta^2} + \alpha} \right\}$$

$$= \sqrt{\alpha^2 + \beta^2} [(\alpha + \beta) \sqrt{\alpha^2 + \beta^2}]$$

$$= (\alpha + \beta)(\alpha^2 + \beta^2)$$

31. $x = n\pi - \tan^{-1} 3$

$$\Rightarrow \tan^{-1} 3 = n\pi - x \Rightarrow \tan(n\pi - x) = 3$$

$$\Rightarrow -\tan x = 3 \Rightarrow \tan x = -3$$

$$\Rightarrow \tan 2x = \frac{2 \tan x}{1 - \tan^2 x} = \frac{3}{4}$$

and

$$\cos x = \pm \frac{1}{\sqrt{1 + \tan^2 x}} = \pm \frac{1}{\sqrt{10}}$$

On substituting these value in the given equation

we find only $\cos x = -\frac{1}{\sqrt{10}}$ satisfies the equation.

So that the given equation holds for values of x for

$$\text{which } \tan x = -3 \text{ and } \cos x = -\frac{1}{\sqrt{10}}$$

Which is possible if x lies in the second quadrant and so n must be odd integer.



LEVEL - IV

1. **Assertion (A) :** If $0 < x < \frac{\pi}{2}$ then

$$\sin^{-1}(\cos x) + \cos^{-1}(\sin x) = \pi - 2x$$

Reason (R) : $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x \quad \forall x \in [0, 1]$

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true but R is not correct explanation of A

3) A is true but R is false

4) A is false but R is true

INVERSE TRIGONOMETRIC FUNCTIONS

2. **Statement -1:** Number of roots of the equation $\cot^{-1} x + \cos^{-1} 2x + \pi = 0$ is zero.

Statement - 2: Range of $\cot^{-1} x$ and $\cos^{-1} x$ is $(0, \pi)$ and $[0, \pi]$ respectively

- 1) Both the statements are true and statement 2 is the correct explanation of statement 1
 2) Both the statements are true but statement 2 is not the correct explanation of statement 1
 3) statement 1 is true and statement 2 is false
 4) statement 1 is false and statement 2 is true

3. Let (x, y) be such that

$$\sin^{-1}(ax) + \cos^{-1}(y) + \cos^{-1}(bxy) = \pi/2$$

- a) If $a = 1, b = 0$, then (x, y) lies on

p) $x^2 + y^2 = 1$

- b) If $a = 1, b = 1$, then q) $(x^2 - 1)(y^2 - 1) = 0$

- c) $a = 1, b = 2$, then r) $y = x$

- d) $a = 2, b = 2$ then s) $(4x^2 - 1)(y^2 - 1)$

1) $a \rightarrow p, b \rightarrow q, c \rightarrow p, d \rightarrow s$

2) $a \rightarrow p, b \rightarrow q, c \rightarrow s, d \rightarrow r$

3) $a \rightarrow r, b \rightarrow q, c \rightarrow s, d \rightarrow r$

4) $a \rightarrow p, b \rightarrow q, c \rightarrow r, d \rightarrow s$

4. Which of the following function(s) is/are identical

(i) $f(x) = \cot(\cot^{-1}x)$; $g(x) = x$

(ii) $f(x) = \sin^{-1}(3x - 4x^3)$; $g(x) = 3\sin^{-1}x$

(iii) $f(x) = \sec^{-1}x + \operatorname{cosec}^{-1}x$; $g(x) = \frac{\pi}{2}$

(iv) $f(x) = \tan(\cot^{-1}x)$; $g(x) = \cot(\tan^{-1}x)$

(v) $f(x) = e^{\ln \cot^{-1}x}$; $g(x) = \cot^{-1}x$

(vi) $f(x) = e^{\ln \sec^{-1}x}$; $g(x) = \sec^{-1}x$

1) only i, iv, v 2) i, iii, iv only

3) ii, v, vi only 4) i, iv only

5. Which of the following function(s) is/are periodic?

(A) $f(x) = \frac{2^x}{2^{[x]}}$ where $[]$ denotes greatest integer function

(B) $g(x) = \operatorname{sgn}\{x\}$ where $\{x\}$ denotes the fractional part function

(C) $h(x) = \sin^{-1}(\cos(x^2))$

(D) $k(x) = \cos^{-1}(\sqrt{\sin x})$

1) only A, B, D 2) only B, D, C

3) only B, A, C 4) only A, D, C

6. Solution of the system of inequalities involving inverse circular functions

MATHS BY Er. RG SIR

$\arctan^2 x - 3 \arctan x + 2 > 0$ and $[\sin^{-1}x] > [\cos^{-1}x]$ where $[]$ denotes the greatest integer function.

1) $x \in (\sin 1, 1)$

2) $x \in [\sin 1, 1]$

3) $x \in [\sin 1, 1)$

4) $x \in (\sin 1, 1]$

LEVEL - IV-KEY

01) 1 02) 1 03) 1 04) 1 05) 1 06) 2

LEVEL - IV-HINTS

1) A: $\sin^{-1}\left(\sin\left(\frac{\pi}{2} - x\right)\right) + \cos^{-1}\left(\cos\left(\frac{\pi}{2} - x\right)\right)$

Reason: $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$ is correct.

3) $\sin^{-1}(ax) + \cos^{-1}y + \cos^{-1}(bxy) = \frac{\pi}{2}$

$$\sin^{-1}(ax) + \cos^{-1}(bxy) = \frac{\pi}{2} - \cos^{-1}y$$

use $\sin^{-1}x + \sin^{-1}y$ for LHS

we get $abx^2y + \sqrt{1-b^2x^2y^2}\sqrt{1-a^2x^2} = y$

$$\Rightarrow abx^2y - y = -\sqrt{1-b^2x^2y^2}\sqrt{1-a^2x^2}$$

squaring on both sides and simplify we get

$$y^2 + a^2x^2 + b^2x^2y^2 - 2abx^2y^2 = 1$$

Now verify the options

5) A) period of $x - [x]$ is 1

B) period of $\{x\}$ is 1, C) period does not exist.

D) period is 2π

6) $(\tan^{-1}x)^2 - 3\tan^{-1}x + 2 > 0$ or

$$t^2 - 3t + 2 > 0, (t-2)(t-1) > 0$$

$(\tan^{-1}x - 2)$ is always < 0

$$\tan^{-1}x < 1 \Rightarrow x < \tan 1, \therefore x \in (-\infty, \tan 1)$$

again $[\sin^{-1}x] > [\cos^{-1}x]$

$[\sin^{-1}x]$ can take the values $\{-2, -1, 0, 1\}$

and $[\cos^{-1}x]$ can take the values $\{0, 1, 2, 3\}$

$\therefore [\sin^{-1}x]$ can be greater than $\cos^{-1}x$ only.

If $[\sin^{-1}x] = 1$ and $[\cos^{-1}x] = 0$

Now $[\sin^{-1}x] = 1, 1 \leq \sin^{-1}x < 2$

$\therefore \sin 1 \leq x \leq 1$ and $[\cos^{-1}x] = 0$

$0 \leq \cos^{-1}x < 1, \cos 1 \leq x \leq 1$

$\therefore$ Now x must satisfy $x \in [\sin 1, 1]$